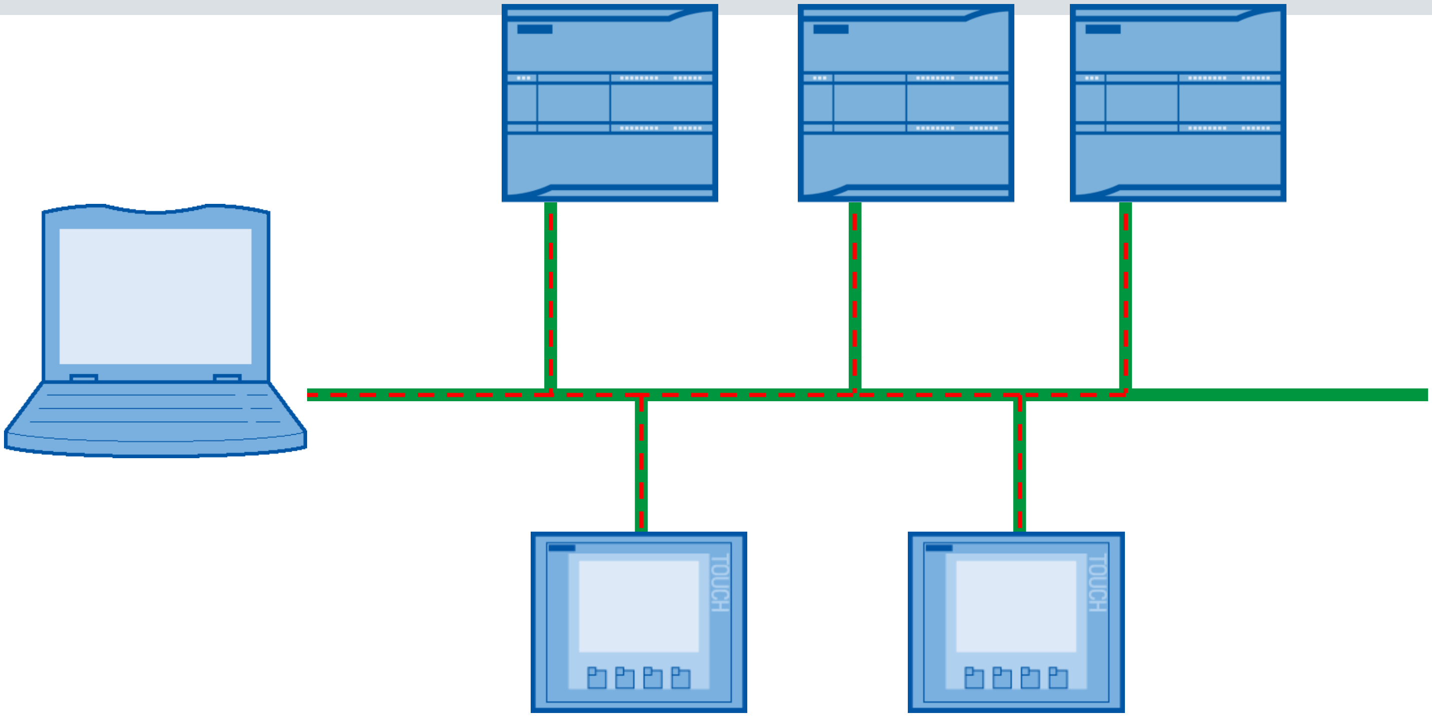
Communication capabilities of the S7-1200, ASi and IO-Link

S7-1200 CPUs use a PROFINET connection to STEP 7 Basic, S7-1200 CPUs, and HMI panels



S7-200 CPUs use an RS485 connection to a PPI network of CPUs and HMI panels.
An expansion Ethernet module must be added for Ethernet communication.

Integrated PROFINET interface

For programming, connection of HMI devices and communication CPU-CPUUp to 16 Ethernet connections possible RJ45 connector with crossover cable detection.

Speed 10/100Mb/s

Supported protocols:

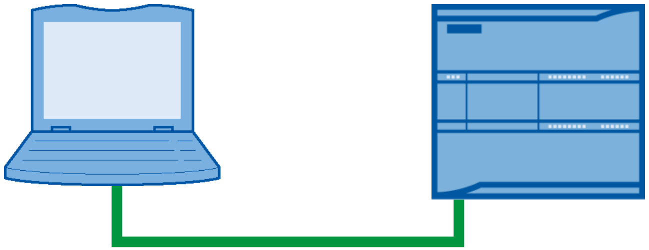
- TCP/IP
- Native ISO to TCP
- S7- communication

Simple creation of communication networks between HMIs and PLCs

S7-1200 integrated PROFINET (Ethernet) - interface

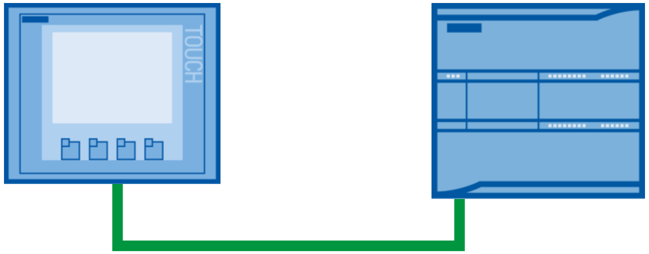
Communication with the STEP 7 Basic software

- CPU hardware configuration
- Project download
- Run-time variable monitoring / modifying
- Run-time Force I/O states
- Diagnostics



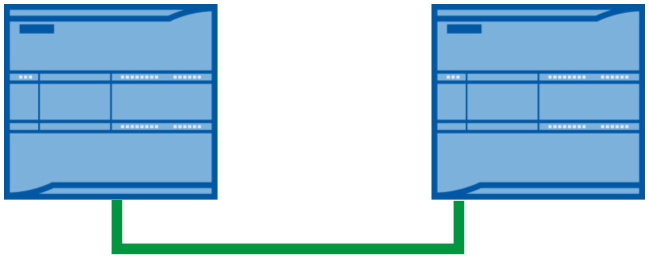
Communication with HMI panels

- Data from / to CPU
- System Diagnostics

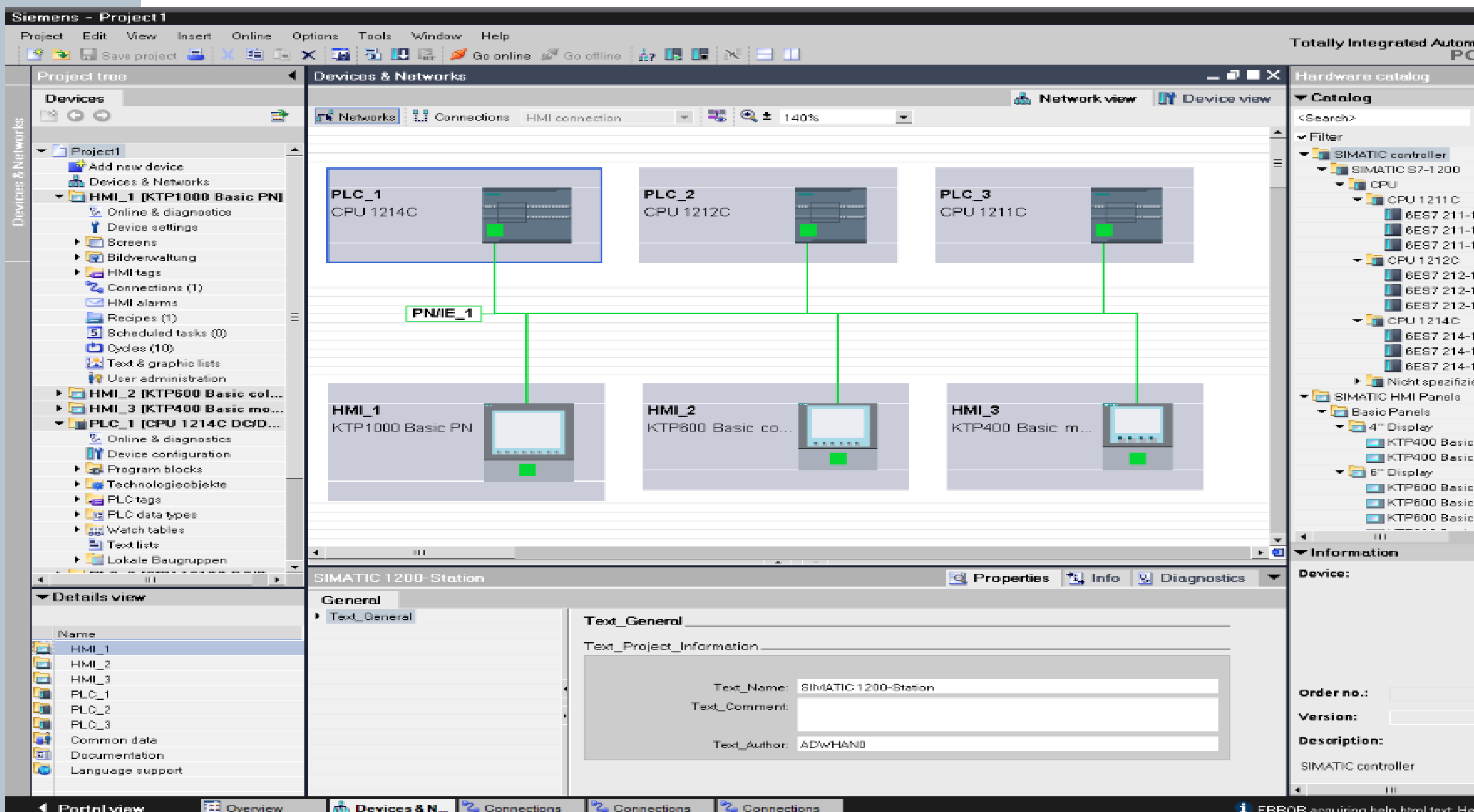


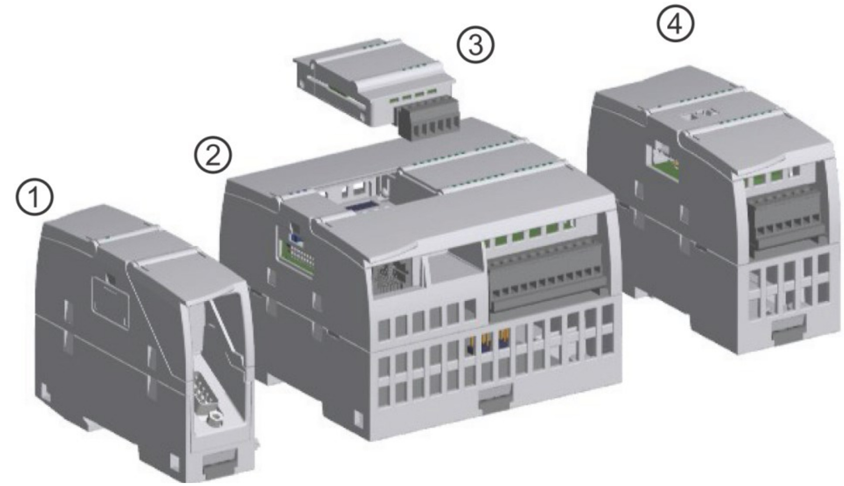
Communication from CPU to CPU

- Open communication with TSEND / TRCV instructions
- Supported Protocols
    - TCP/IP native
    - ISO on TCP
- S7-communication (PUT / GET) server only



Integrated PROFINET interface





Module	Type	Description
① Communication module (CM)	RS232	Full-duplex
	RS422/485	Full-duplex (RS422) Half-duplex (RS485)
	PROFIBUS Master	DPV1
	PROFIBUS Slave	DPV1
	AS-i Master (CM 1243-2)	AS-Interface
① Communication processor (CP)	Modem connectivity	GPRS
① Communication board (CB)	RS485	Half-duplex
① TeleService	TS Adapter IE Basic[1]	Connection to CPU
	TS Adapter GSM	GSM/GPRS
	TS Adapter Modem	Modem
	TS Adapter ISDN	ISDN
	TS Adapter RS232	RS232

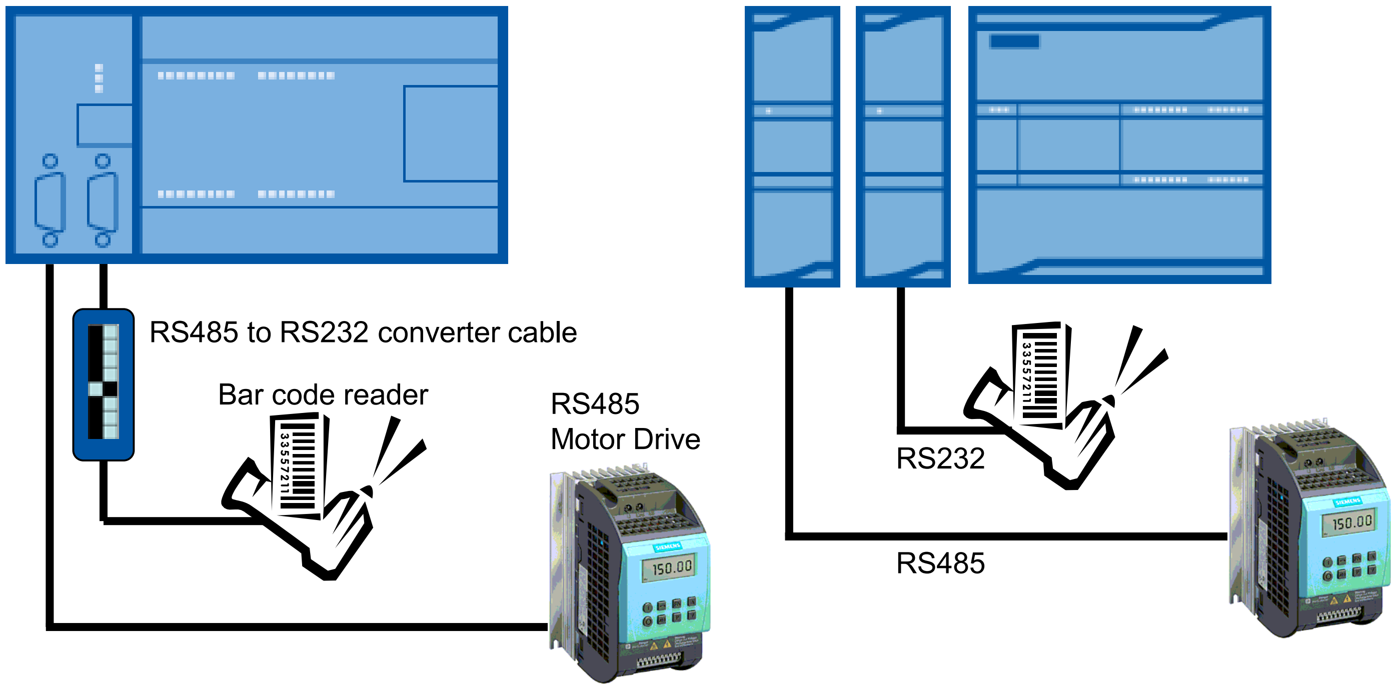
Serial communication for S7-200 and S7-1200

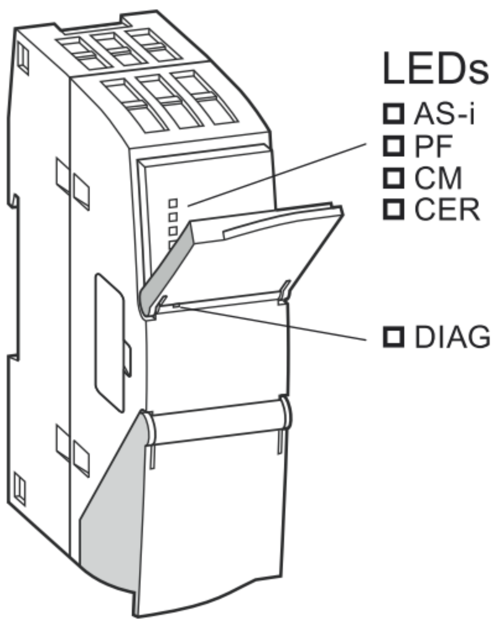
S7-1200 CPU communication via RS232 and RS485 connections

- ASCII protocol (character based serial communication) uses STEP 7 Basic point-to-point (PtP) instructions
- USS Drive protocol is programmed with STEP 7 Basic USS Library instructions
- MODBUS protocol is programmed with STEP 7 Basic MODBUS Library instructions

S7-200 CPUs have 1 or 2 on-board RS485 serial connections

S7-1200 CPUs have 1 on-board PROFINET (ETHERNET) connection. Use the RS232 and RS485 modules for PtP communication





AS-i	AS-i interface
PF	Peripheral fault
CM	Configuration mode
CER	Error in configuration
DIAG	Diagnosis

AS-Interface: The simple system solution

Users Guideline (GB)

AS-International Association

The perfect solution

AS-Interface

- Optimised for the connection of Sensors and Aktuators
- Cost effective
- Operation in difficult environmental conditions
- Reliable and safe
- Real time capable
- Applicable universal
- Easy to install
- Easy to extend
- Open to higher levels
- Safe against interference
- Standardized EN 50295, IEC62026

System Facts

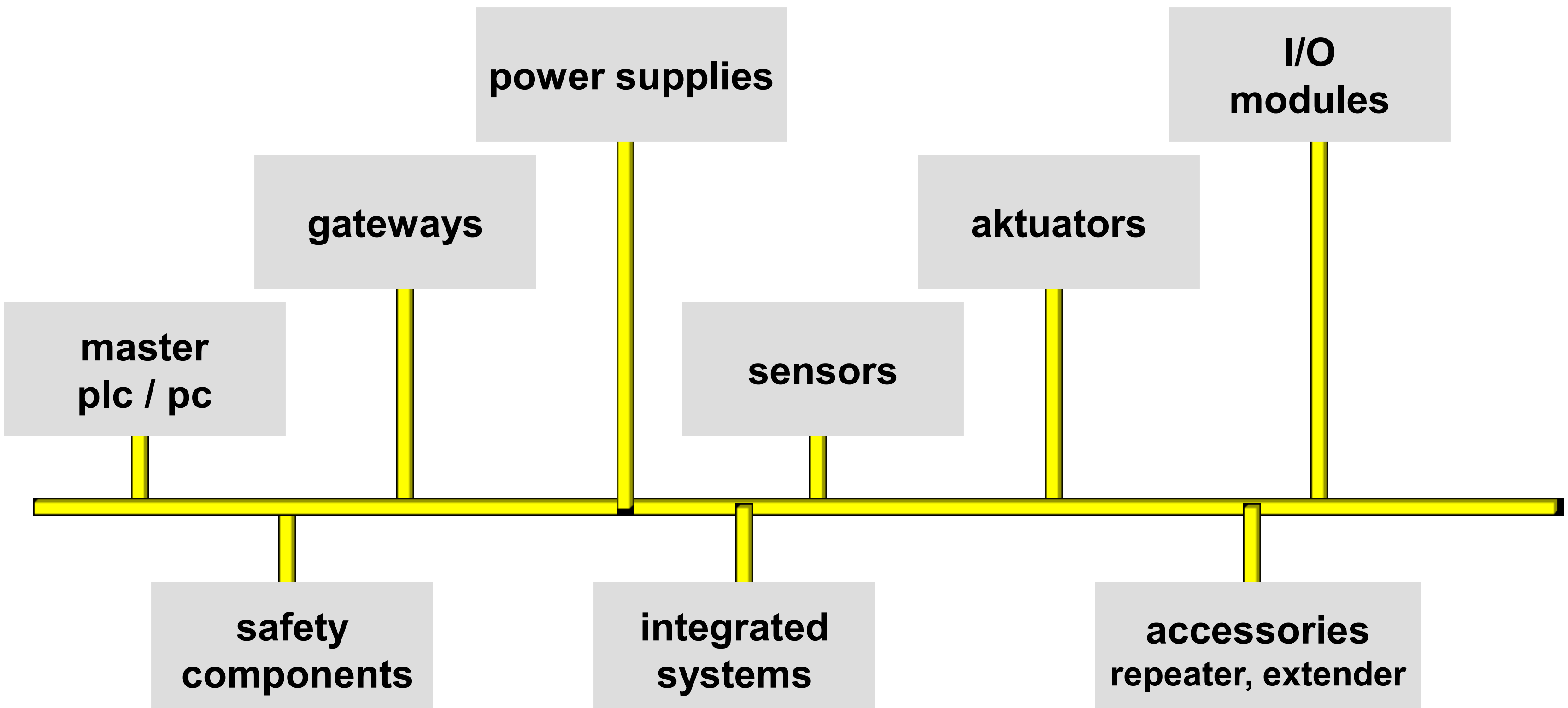
- In field use (IP 65/67) as well as in the switch cabinet
- piercing technologies
- cable length 100 m, extendable via Repeater up to 500m
- reliable data transmission
- easy commissioning
- address to be set via bus connector

Facts about AS-Interface

- master-slave principle
- no limits to the structure
- 2 wire cable for data and power up to 8 A
- Unshielded cable 2 x 1,5mm²
- safe transmission
- 4 digital inputs + 4 digital outputs
- analog value transmission

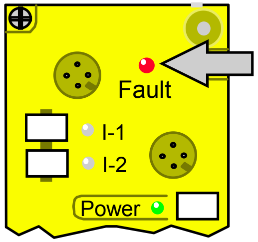
System solution

Product range

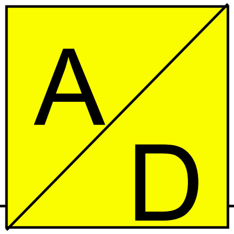


AS-Interface

- 62 Slaves at one Master / A- und B-Slaves
- full downwards compatibility
  - existing applications can be extended by new modules
  - existing Slaves communicate with new masters



- exact diagnostic
- separated detection of configuration and device errors
  - short circuit, overload, missing auxiliary power, communication error
- detailed diagnostics and easy maintenance



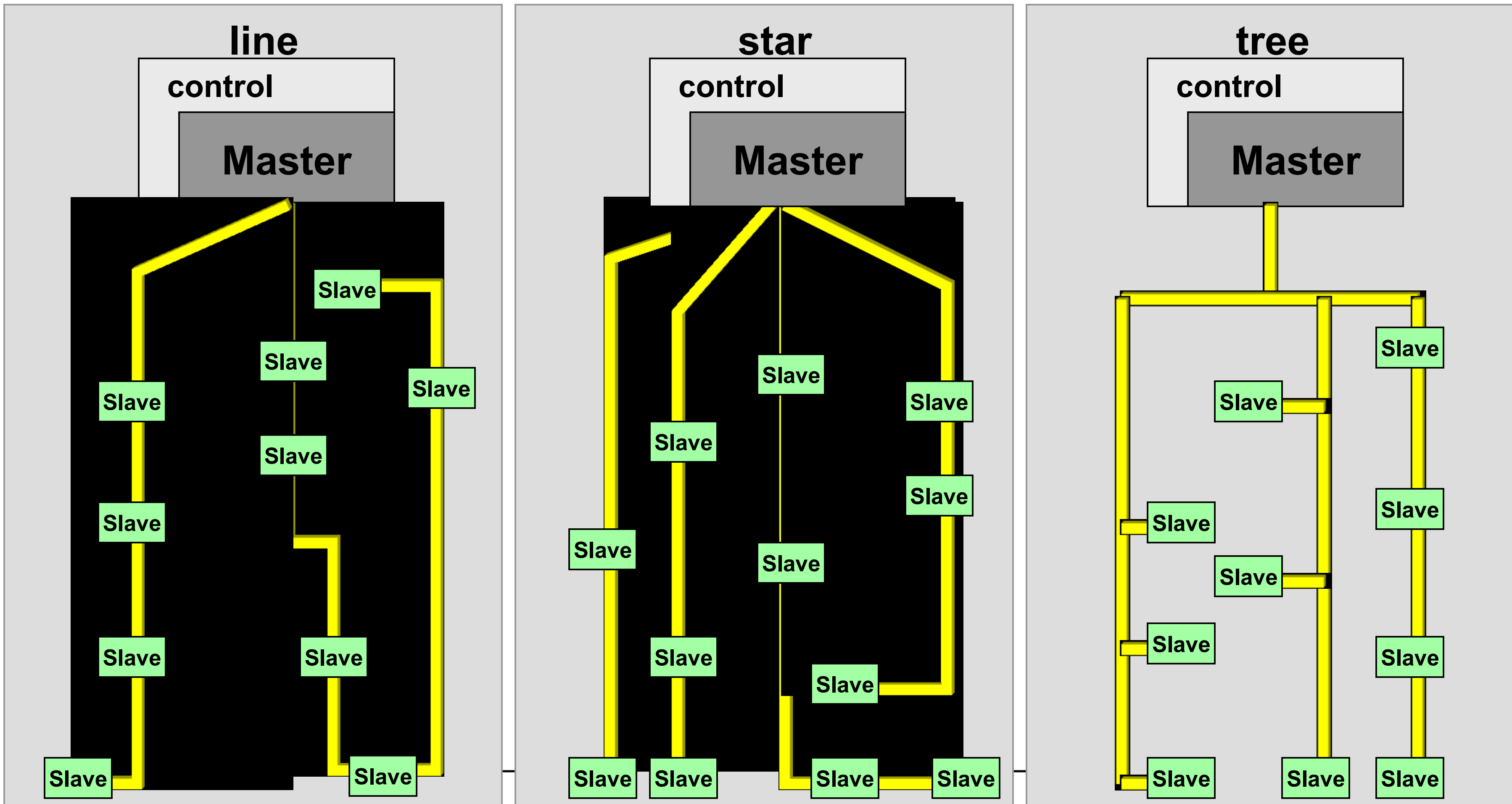
- transmission of analog values
- no unreasonable time delay
- automatic detection of analog participants
- no need for configuration software

System Facts

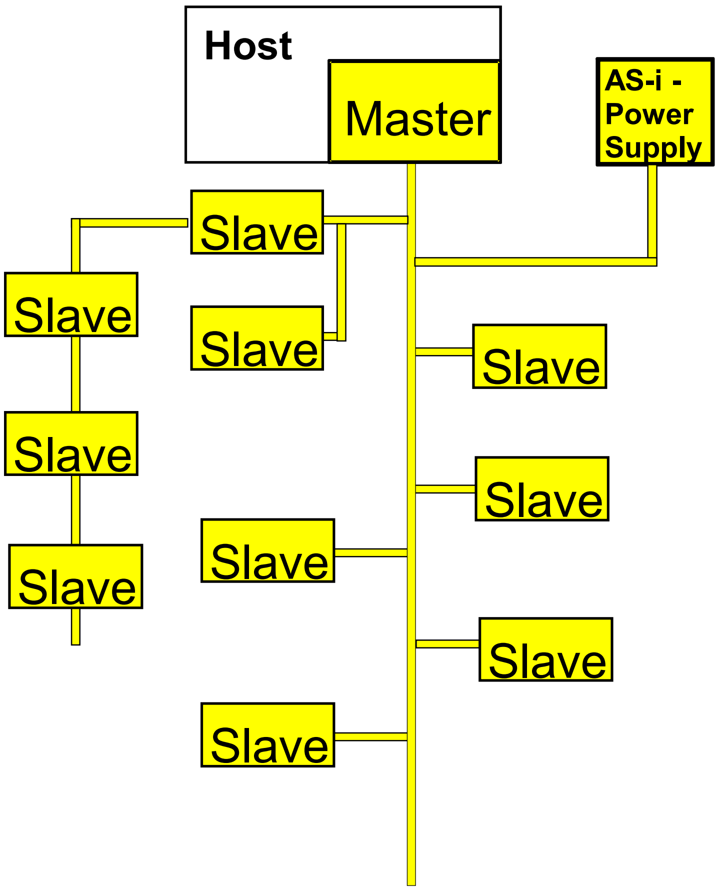
	AS-Interface	Enhanced
number of Slaves	31	62
number of I/O´s	124 I +124 O	248 I + 186 O
max. cycle time	5 ms	using A + B Slaves max.10 ms
analog value transmission	via function block	integrated in the Master up to 124 analog values

Network Topology

no limits to the structure :



System Survey

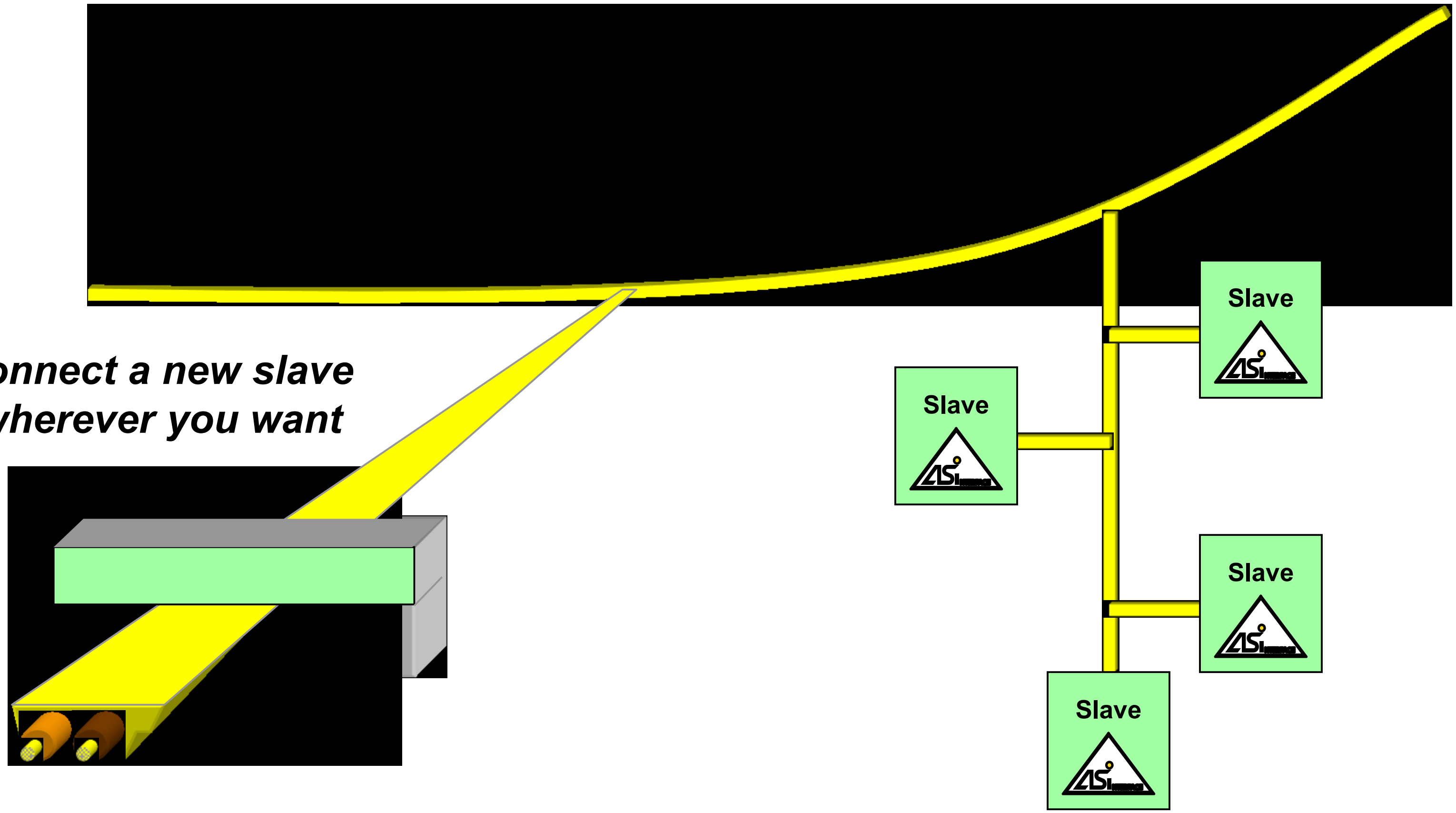


- Single master-slave principle
- Up to 31 slaves with one master
- Per slave up to 4 digital inputs + 4 digital outputs
- Additional 4 parameterbits / slave
- Max. 248 digital in- and outputs
- Also possible: analogue I/O
- Electronic slave addressing
- Free structure of the network

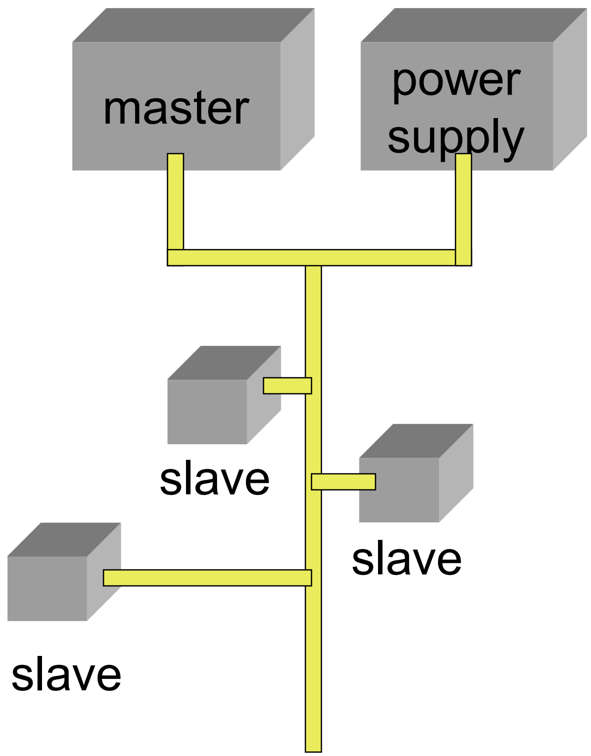
Modular Construction

easy extendable

Connect a new slave wherever you want

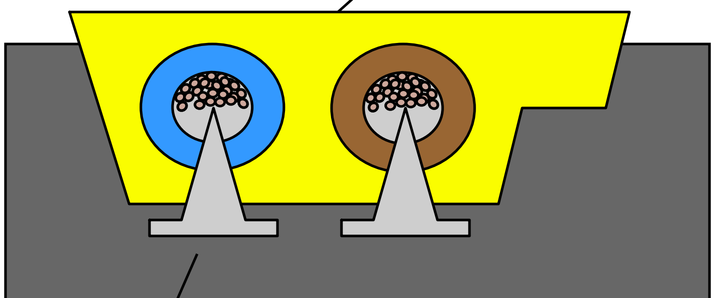


Minimum equipment for using AS-Interface ®



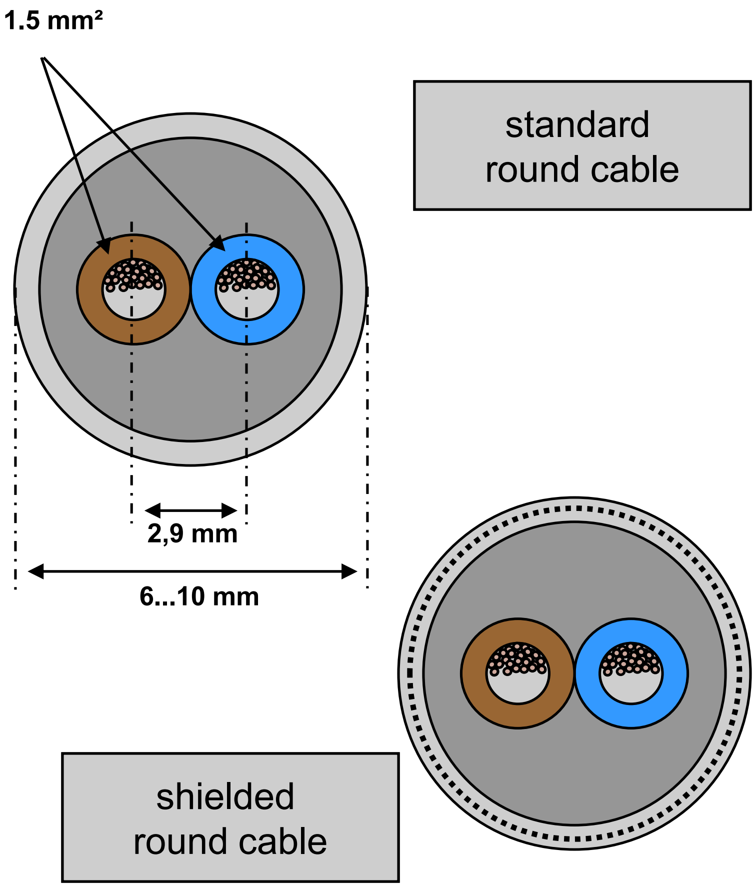
- **Master:**
  PLC Master or Gateway
- **AS-i Power supply:**
  Supply 30,5 VDC with
  integrated data decoupling
- **Cable**
- **Slaves**
  Modules with electrical I/O or
  Sensors with integrated AS-i chip or
  Actuators with integrated AS-i chip

Selecting the correct cable



- **Flat cable**
- **Main benefits**
- Easy and quick installation
  (no cutting, no stripping)
- No wrong installation caused by reversed polarity
- Contact cycles: minimum 5 times
- Unshielded 2 wire cable for data and energy (24 V DC / typical up to 8 A for the bus, higher values allowed)
- Different colours for different voltages
  yellow: AS-Interface:
  black: additional power supply 24V
  red: additional power supply 230 V
- Different cable qualities
  PUR, TPE, EPDM (rubber)

Selecting the correct cable

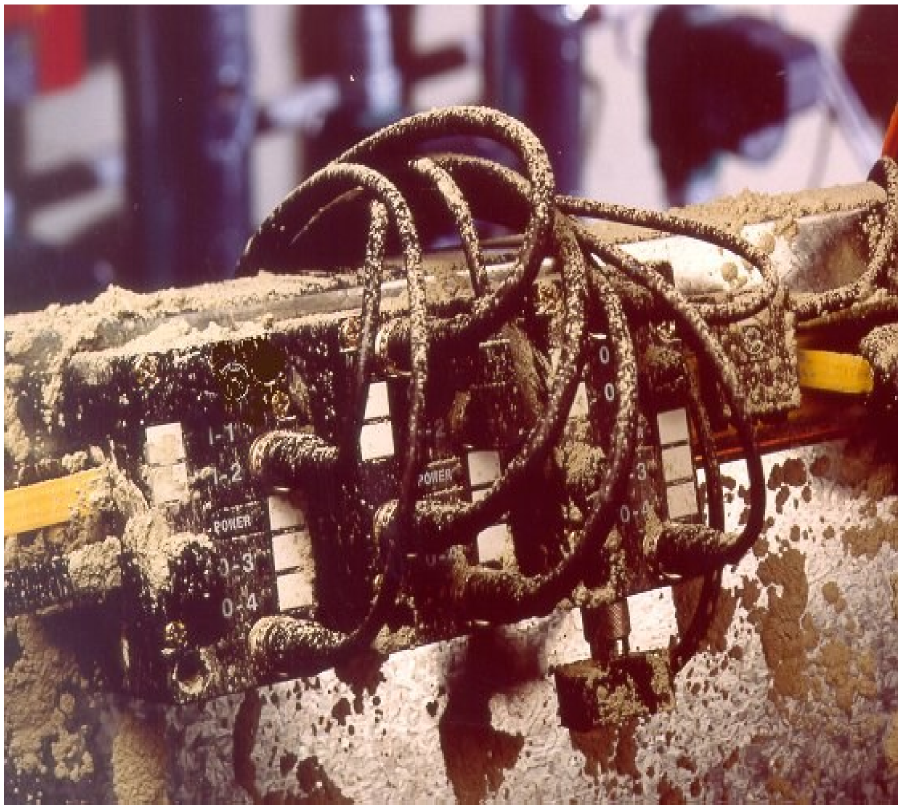


Round cable

Main benefits

- nearly all standard cables can be used
- for installations with special requirements (e.g. highest flexibility in robot applications, trailing chains)
- torsion proof
- connection by screw terminal block
- no cable-loops necessary to get the right orientation by connecting both wires
- shielded versions available (if required)
- cable parameters influence the max. system extension (i.e. capacity)

Protection classes



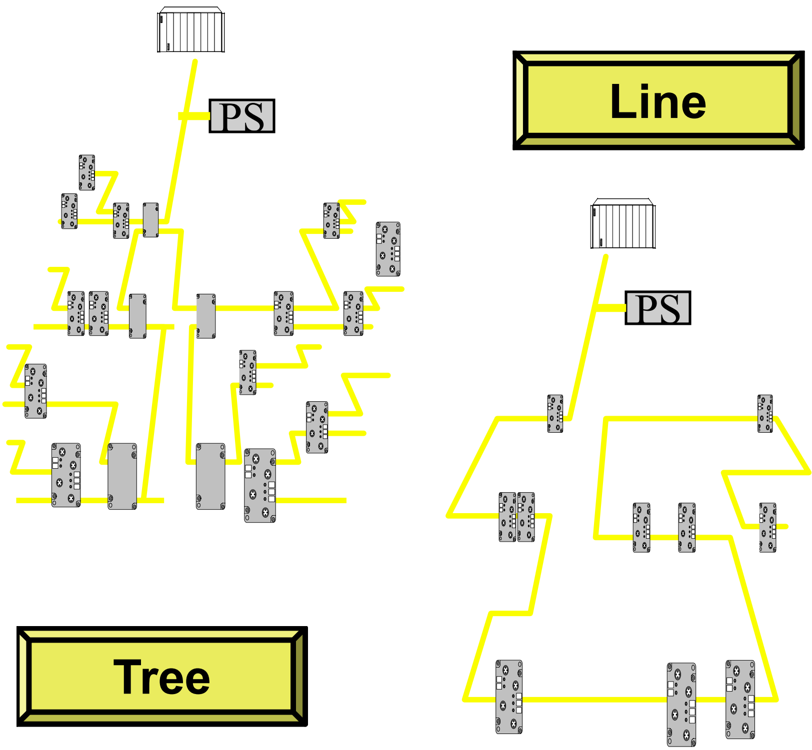
IP67 I/O-modules

- Installed direct in the process (placed near to sensors, actuators)
- No need for additional housings in industrial environment
- Standardised connection by M12-plug or PG-cable glands
- Standardised interface between user module and coupling module
- Support completely the idea of decentralised I/Os

IP20 I/O-modules

- Slim housings for use in cabinets
- In machine installations: need for an additional housings (cabinet, ...)
- Flexible installation (typical: clamps)

Cable lengths and current



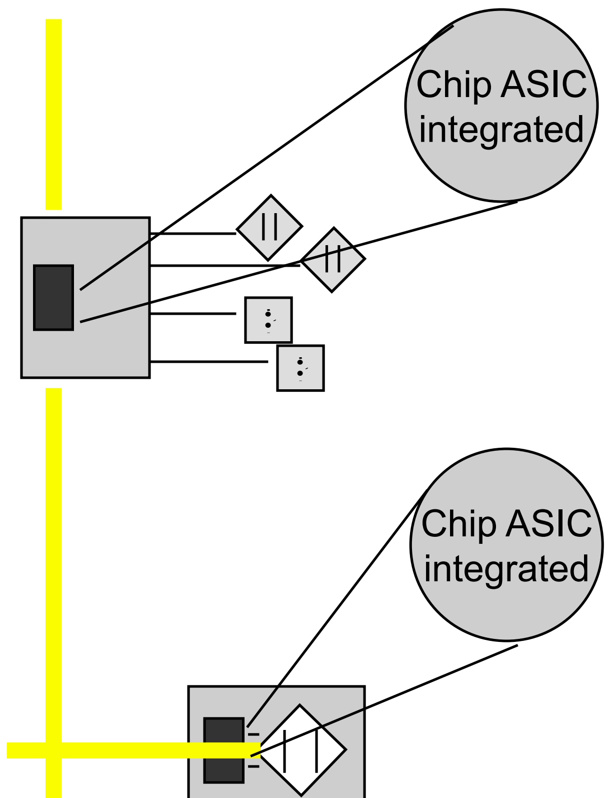
System limits

- 100 m in total (all cable pieces)
- Add also all cable length for connection to the AS-i integrated slaves.
- Allowed topologies: Tree, star, line ...

Total current, voltage drop

- AS-Interface network is typically rated up to 8A (depending on the power supply)
- Max. allowed voltage drop on a whole 100 m-cable: **3 V**
- Operating voltage of slaves: 26,5..31,6 V
- Typical current: 200 mA per slave
- If higher current demanded -> install an additional power supply (black cable)
- Locate the power supply where the highest power demand exists

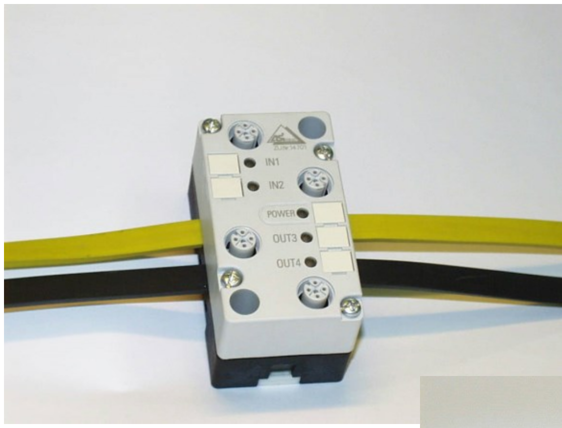
Where is the AS-Interface® chip located ?

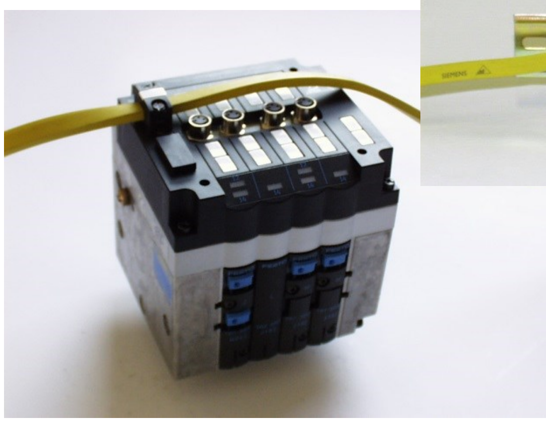


- **The chip** (ASIC) **is:**
- **integrated into an active module with electrical I/O**
  (for connection of standard actuators and sensors)
- **integrated directly into sensors or actuators**

ASIC:
"**A**pplication **S**pecific **I**ntegrated **C**ircuit"

Connection of actuators





Actuators without AS-i

Connection via I/O module

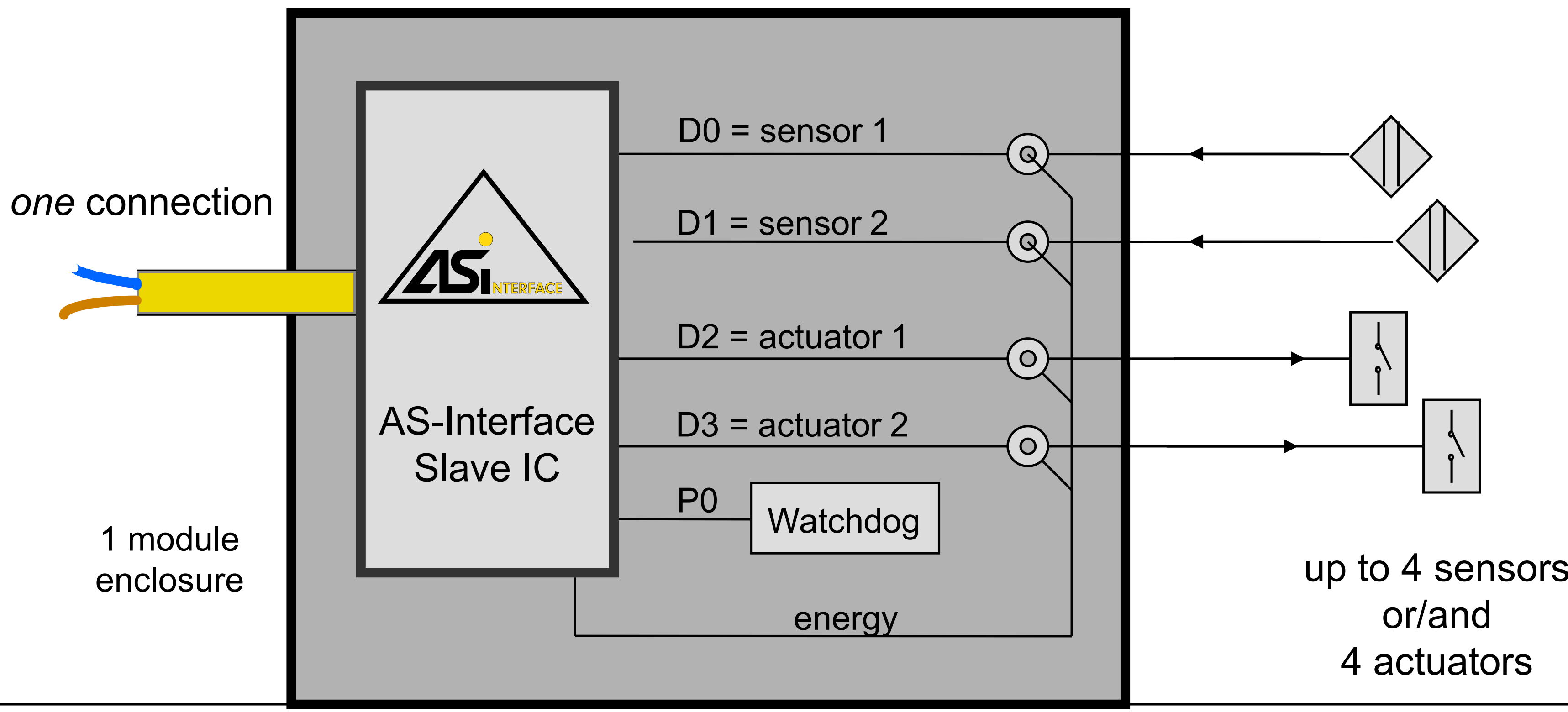
- Valves
- Indication units
- Contactors

Actuators with integrated AS-i

Connection directly on AS-i Cable

- Airbox
- Valve plug
- Valve terminals
- Motor starter
- Indication units
- Electric, pneumatic, hydraulic drives

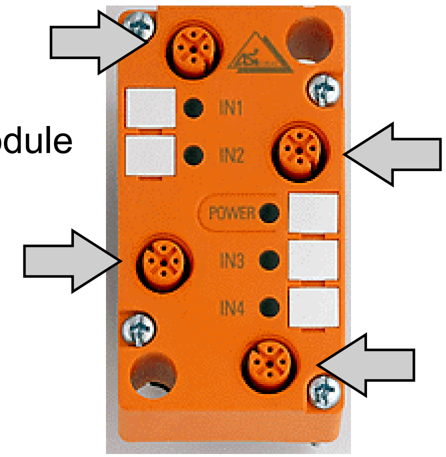
AS-Interface modules allow for connecting conventional sensors and actuators:



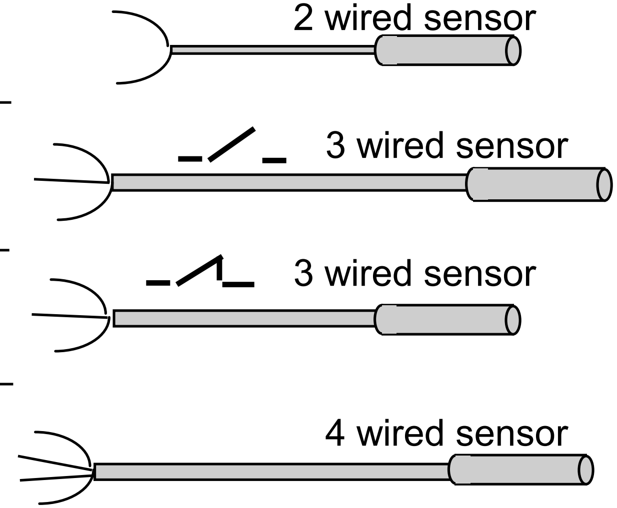
How to connect sensors without own AS-i Interface

pnp or npn

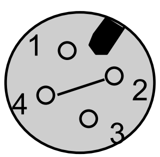
AS-i module



pin	npn	pnp	coulor
	3	1	bn
	4	4	bu
	1	1	bn
	4	4	bl
	3	3	bu
	1	1	bn
	2	2	bl
	3	3	bu
	1	1	bn
	4	4	bl
	2	2	wh
	3	3	bu

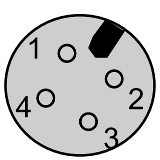


standard-
input
pnp/npn



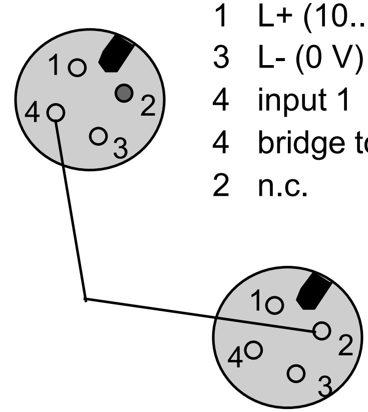
1 L+ (10...24 V)
3 L- (0 V)
4 input
2 bridge to pin 4

double-
input



1 L+ (10...24 V)
3 L- (0 V)
4 input 1
2 input 2

bridged
combi-
input

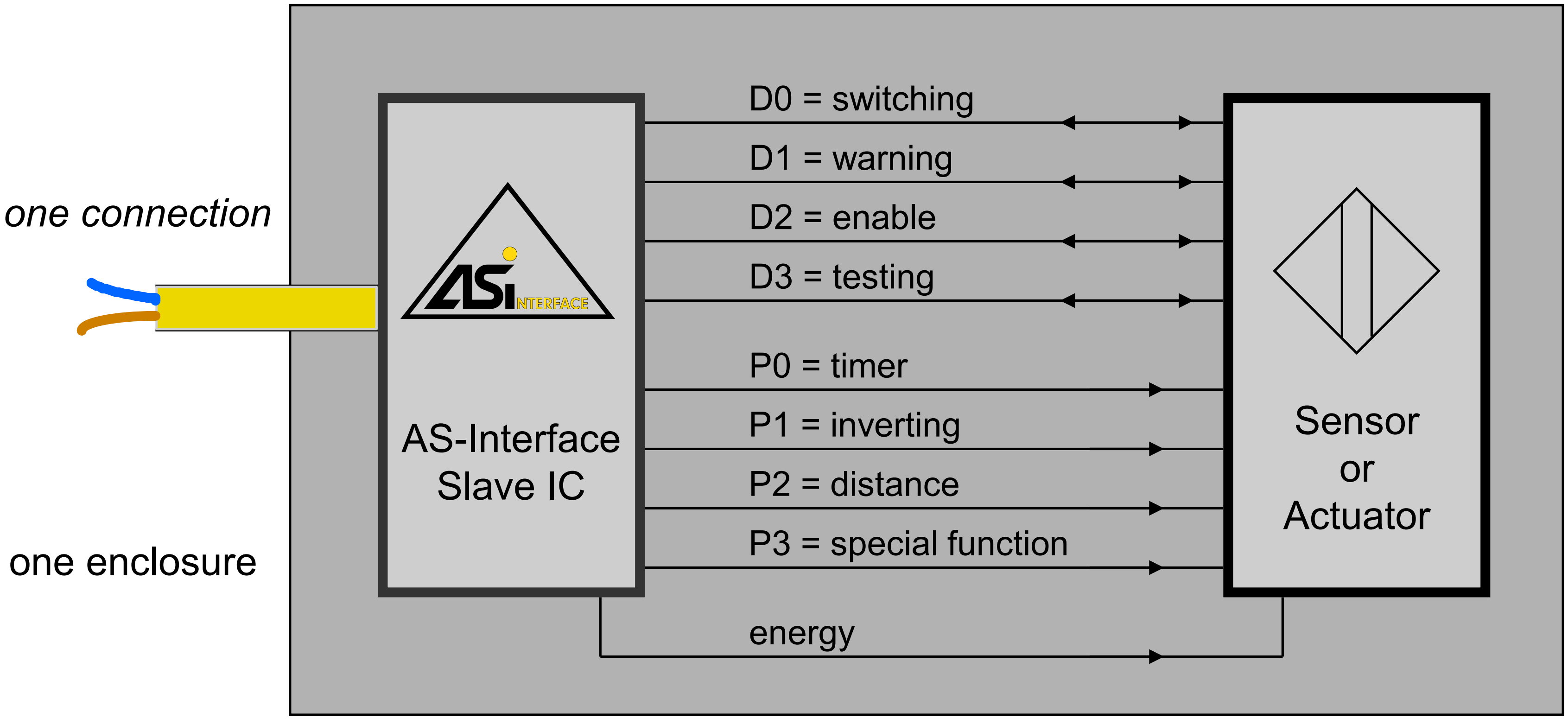


1 L+ (10...24 V)
3 L- (0 V)
4 input 1
4 bridge to socket 2, pin 2
2 n.c.

1 L+ (10...24 V)
3 L- (0 V)
4 input 2
2 bridge to socket 1

System Solution

AS-Interface integrates slave functionality:



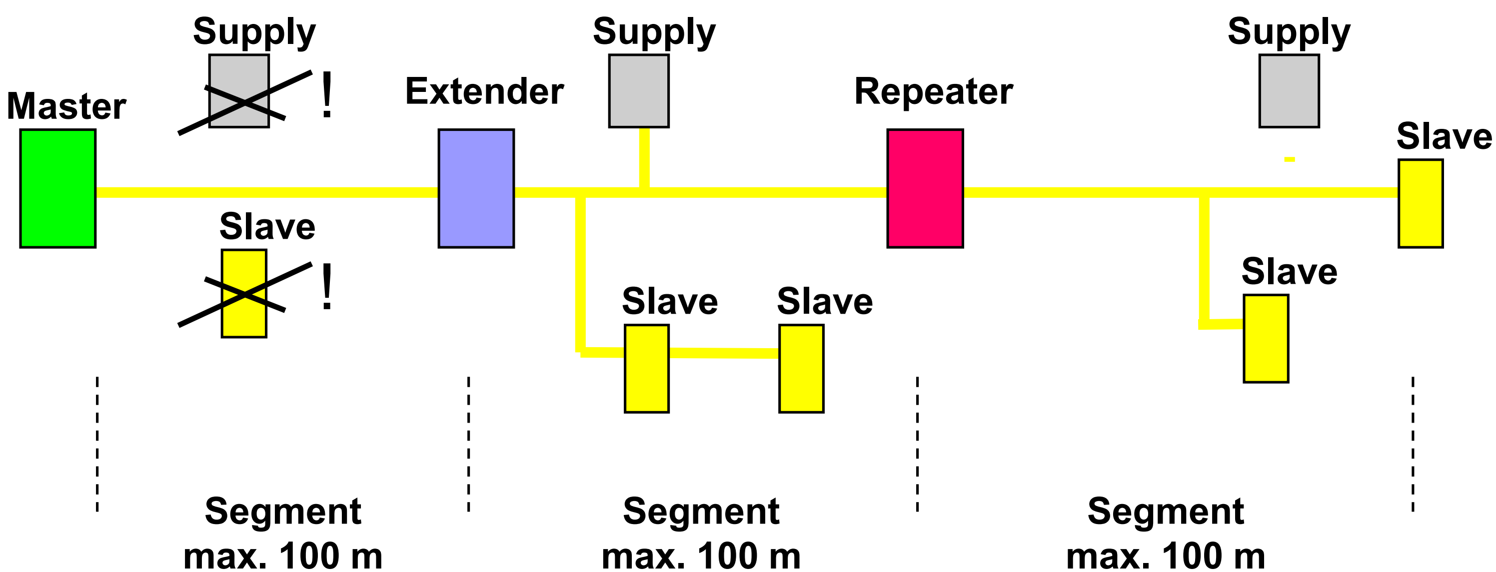
- Slaves with additional functions, like parametrising.
- diagnostics of the net up to the slave
- actuators in IP67 switching in field, not in cabinets

Tips for extending the length of the network

Length of all AS-i cables in a network / segment is up to a maximum of 100 m

It is possible to extend a network up to 300 m (3 segments) by use of an extender and / or repeater:

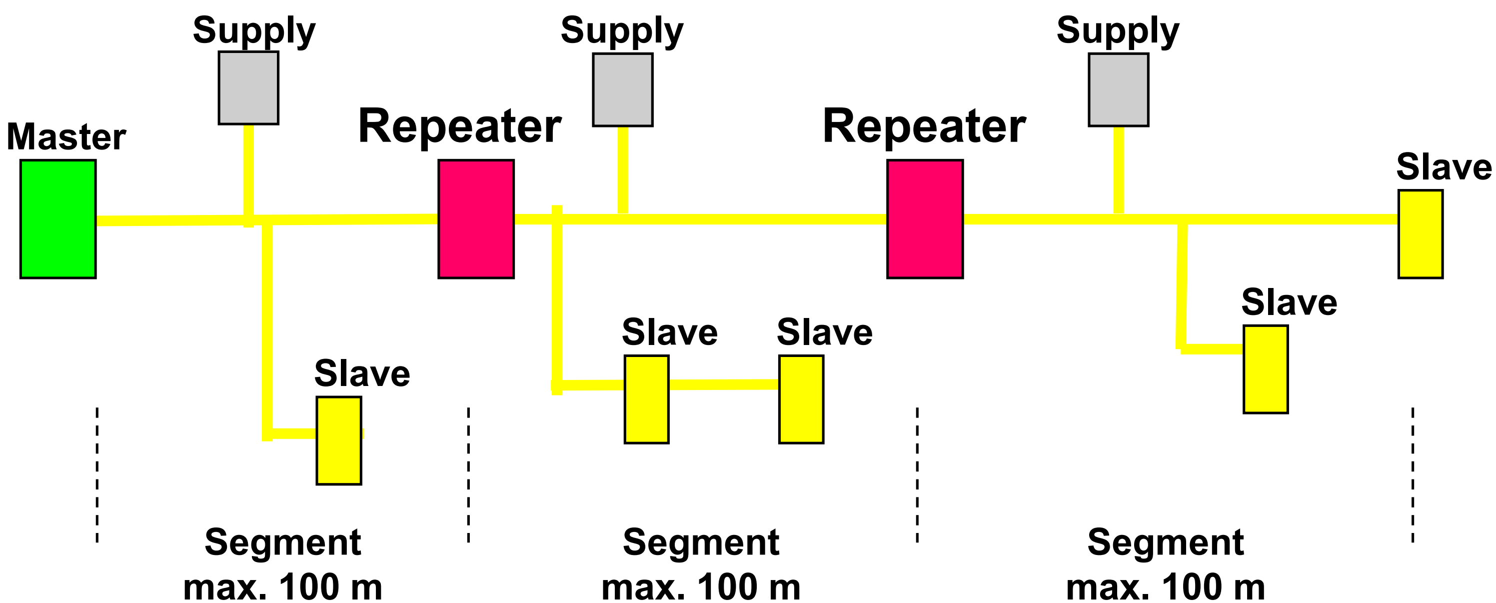
Solution A: *1 extender and 1 repeater*



Max. number of slaves over all is 31 !

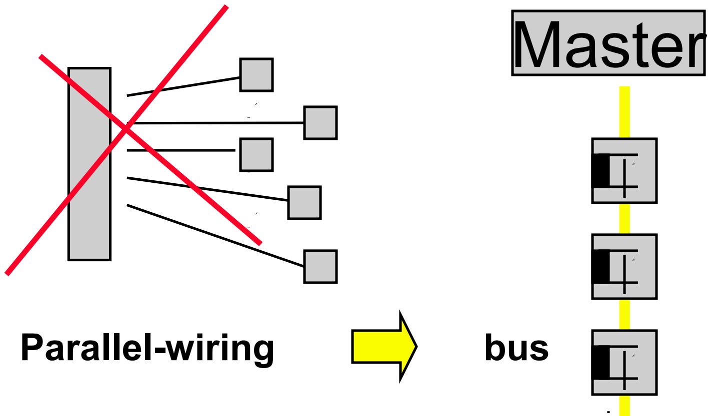
Tips for extending the length of the network

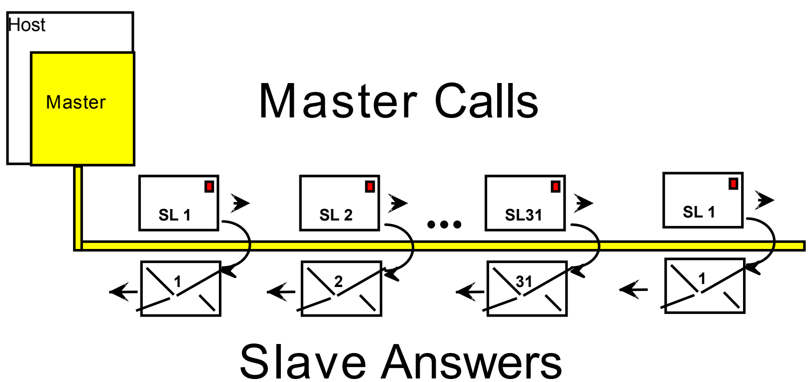
Solution B: *2 repeater*



Max. number of slaves over all is 31 !

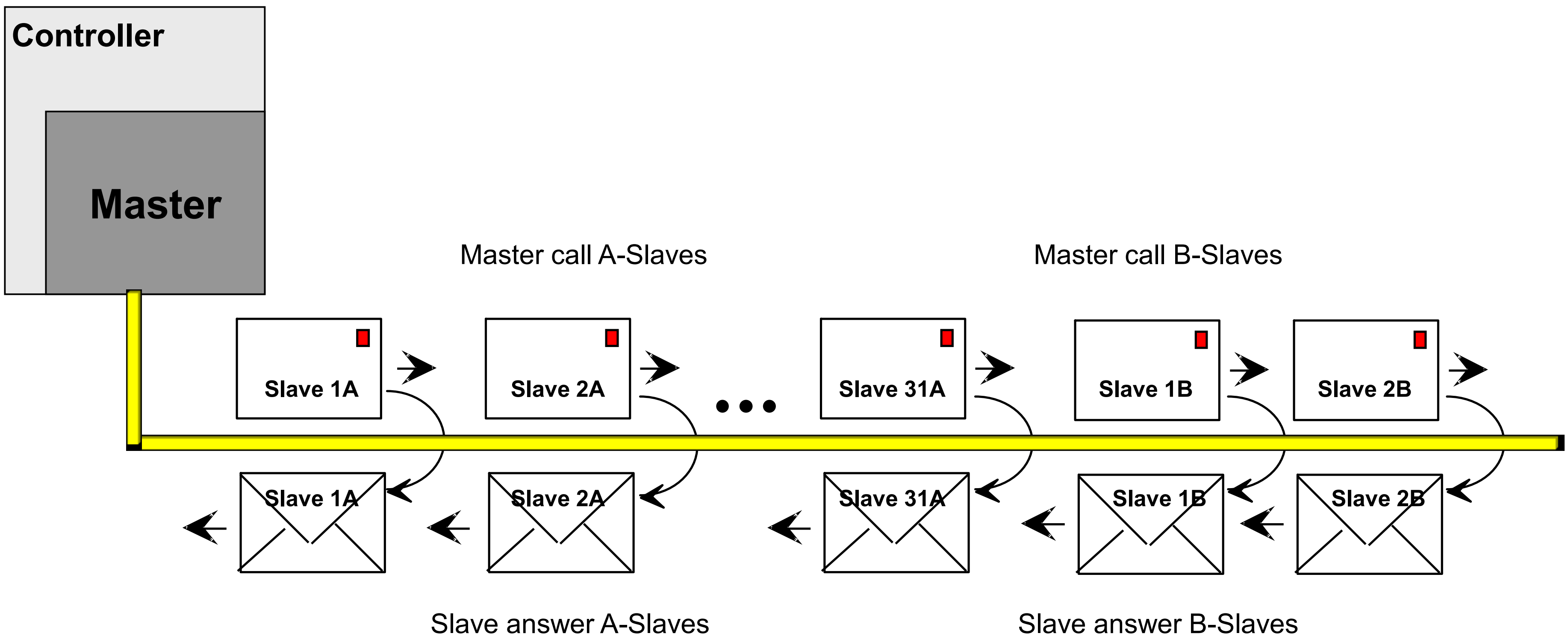
How AS-Interface® works





- **AS-Interface®**
  - A bus system, which substitutes parallel wired installation from plc to sensors and actuators
- **Data and energy on the same cable**
- **1 Master and max. 31 Slaves**
- **Master - slave principle:**
  - The master calls and the called slave answers immediately
- **Total cycle time < 5 ms**
  - with max. numbers of 31 slaves

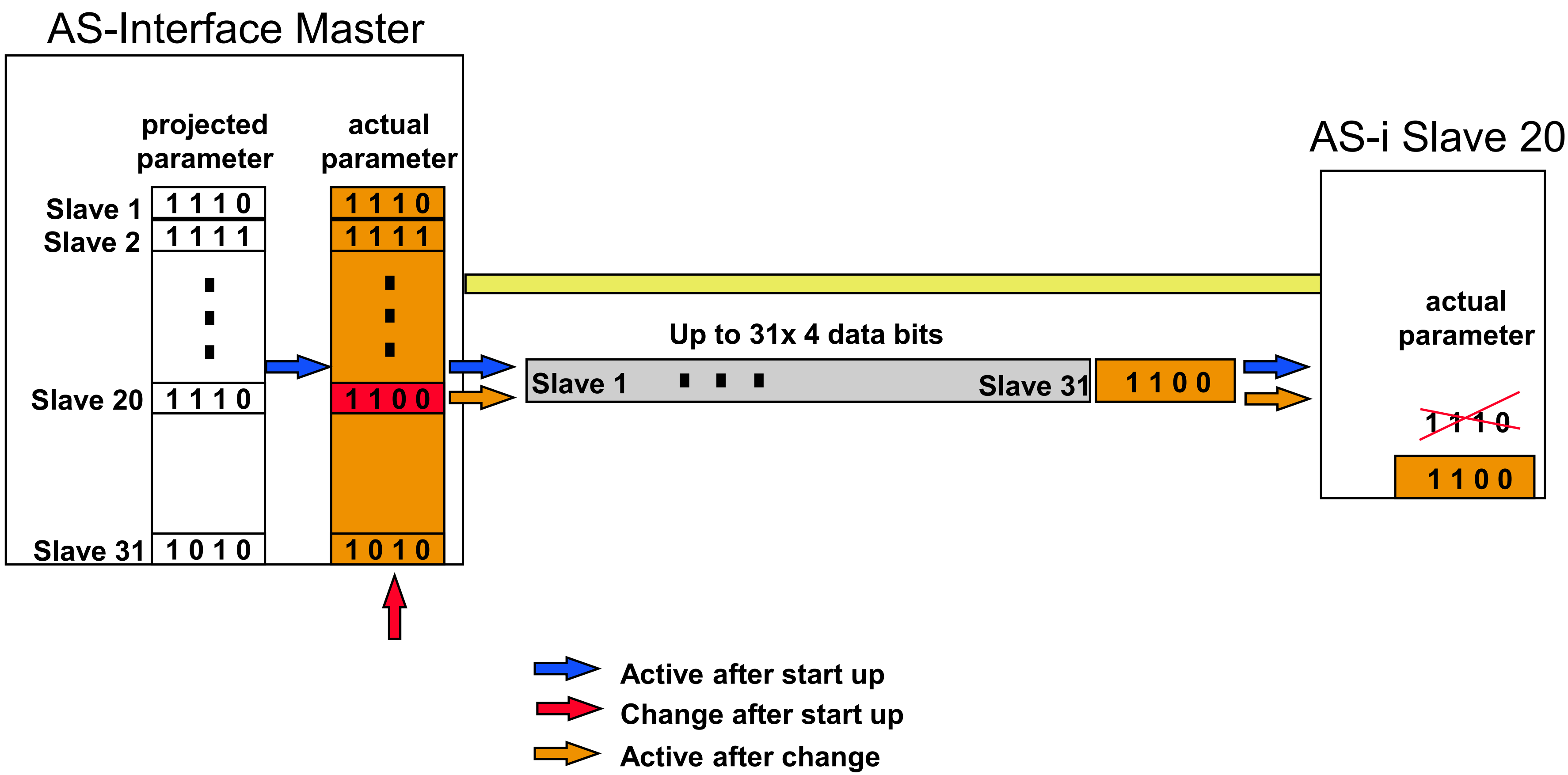
A/B-Slaves functionality



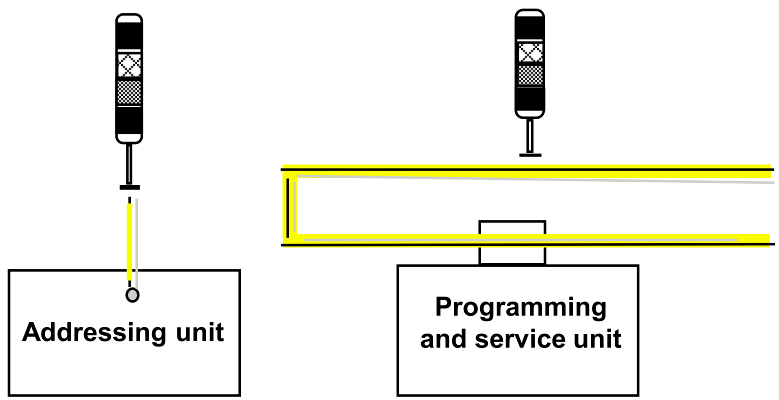
Transmisja

		Master call												Master - pause			Slave answer					Slave pause
0	CB	A4	A3	A2	A1	A0	I4	I3	I2	I1	I0	PB	1		0	I3	I2	I1	I0	PB	1	
ST													EB		ST						EB	

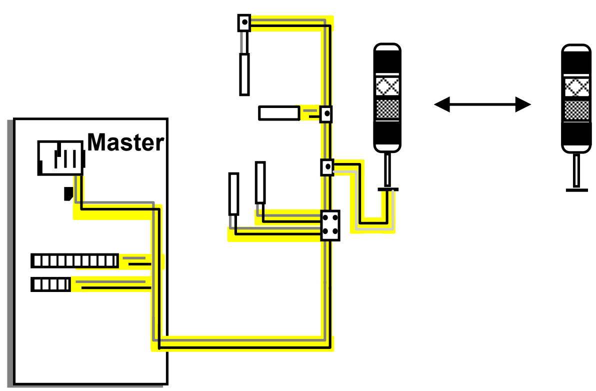
How AS-Interface work with parameter



Addressing

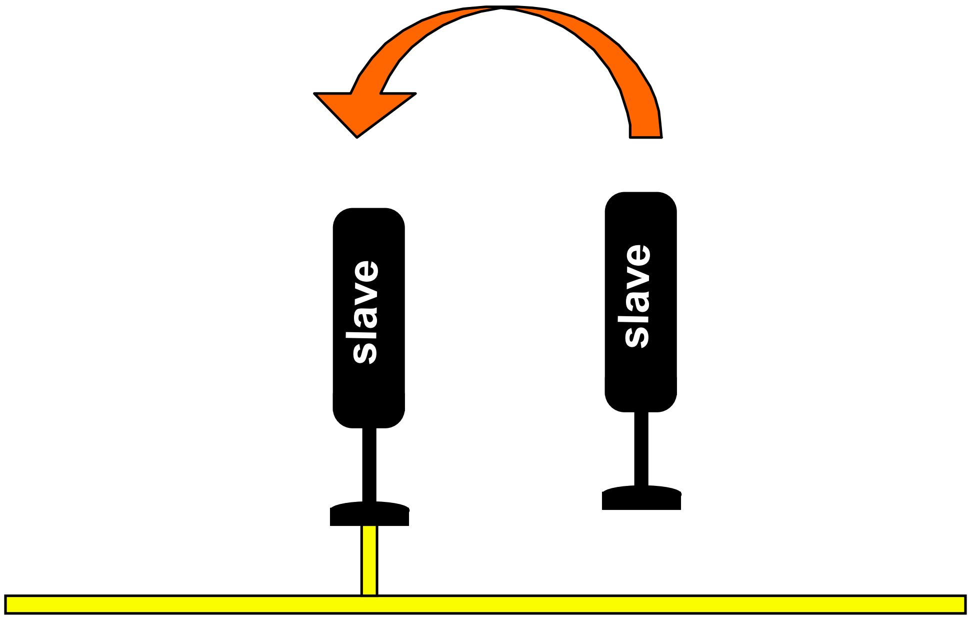


- Single addressing by an addressing unit



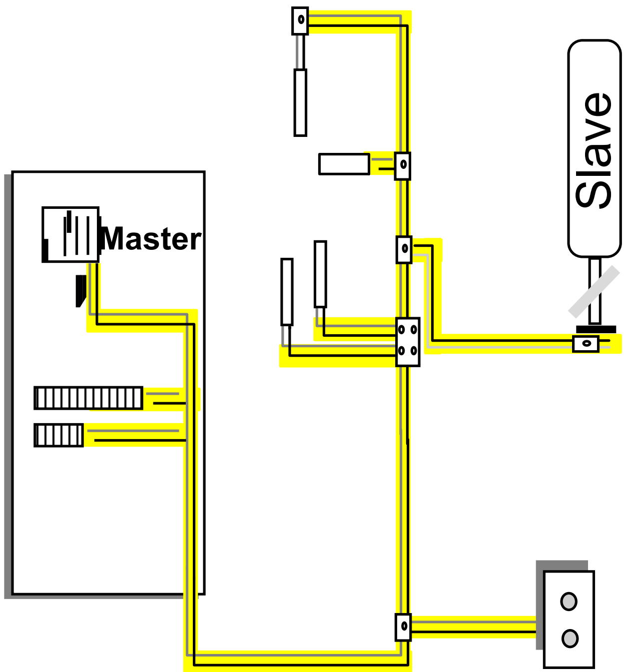
- Automatic addressing by the master
(One by one !)

Replacement of Slaves



- There have to be the same of old and new slave:
- I/O-Code
- ID-Code
- Slave address

No answer of certain slave, why?



- Slave is not yet addressed
- Slave is configured wrong (see I/O table)
- Slave is faulty
- Cable fracture
- No connection
- Address is used twice

Diagnostic methods

LEDs indicate the status of the slaves:

Symptom	Flags on Master		Indication on Slave				Possible cause
			Standard		Enhanced		
	Config Error	Periph. Fault	normal	dual LED	normal	dual LED	
Normal operation	reset	reset	green	green	green	green	
No data exchange	set	reset	green red	rot	green red	red	Master in STOP mode Slave not in LPS Slave with wrong IO/ID RESET on slave aktiv
No data exchange (Address = 0)	set	reset			green red	red/ yellow	Slave Address = 0
Periphery Fault	reset	set	green red	rot/ grün	green red	red/ yellow	To be defined by manufacturer
Serious Periphery ault wit RESET	set	set			green red	red	To be defined by manufacturer

Watchdog

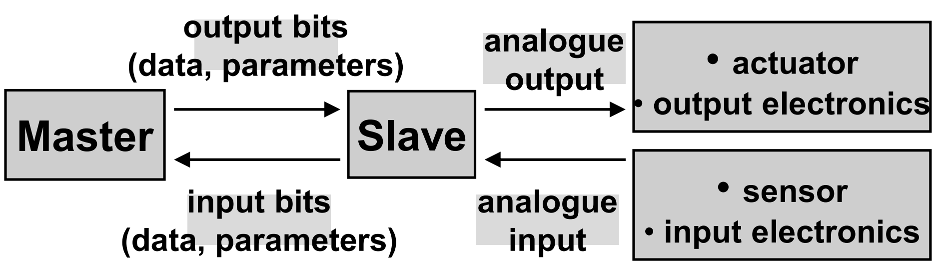
- There are modules with and without watchdog
- The watchdog in a slave monitors the telegrams of master and slaves
- The watchdog is triggered if no master telegrams run for more than 40 ms
- The outputs of the slaves will be switched off

- There is also a special designed watchdog module which simulates short circuit for restet function of the slaves in case of missing communication.

- Possible causes to trigger a watchdog are:

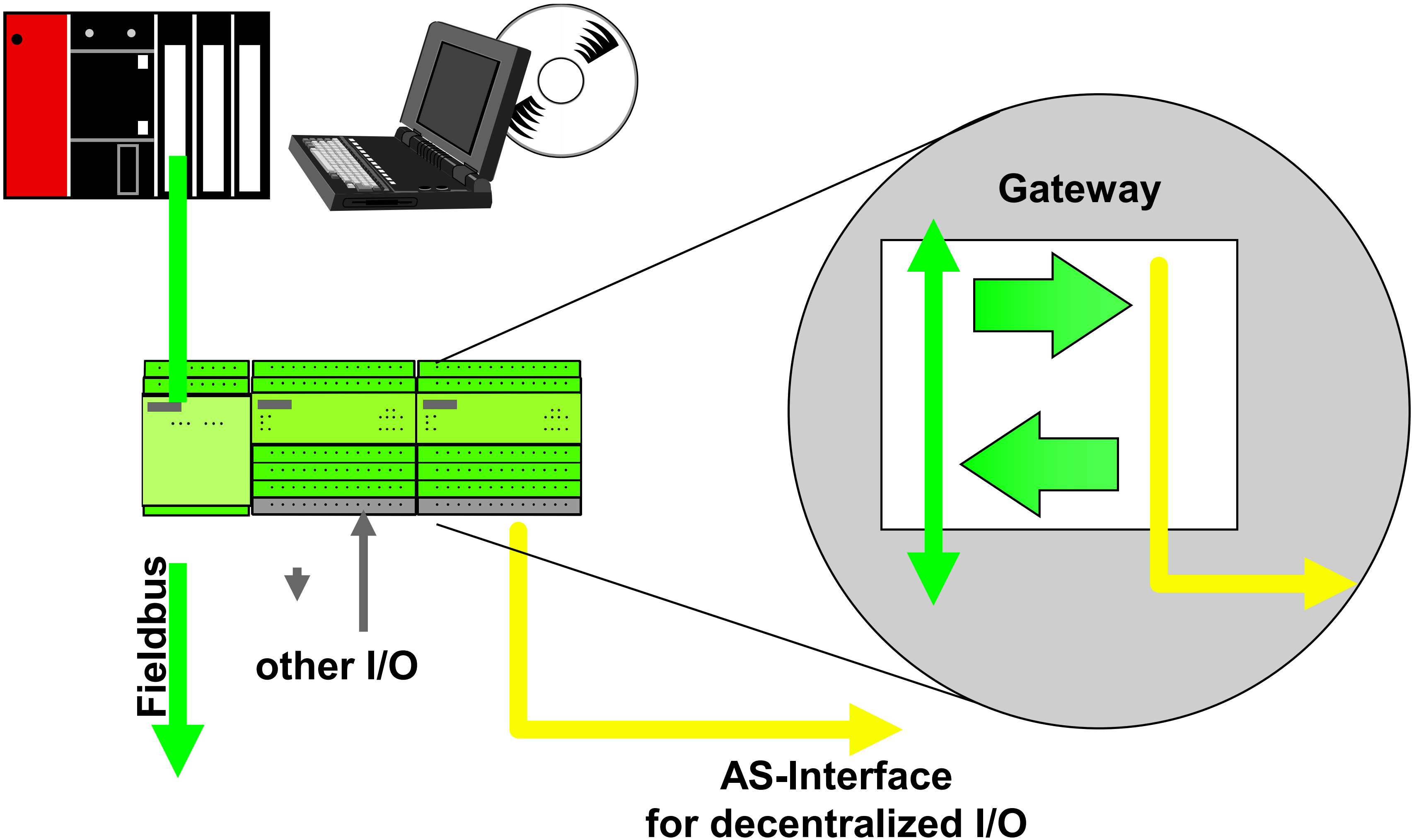
  - AS-Interface cable is broken
  - Master fault
  - Master in stop mode
  - Slave not in list of projected slaves (LPS)

Analogue I/O with AS-Interface



- AS-Interface modules and sensors:
  1. data input (e.g. for measuring temperatures or pressure)
  2. data output as analogue value
- I/O: 0...20 mA, 4...20 mA or 0...10 V
- The data are transferred in parts of 4 bit
- Additional information bits
  S : sign, O : overflow, V : valid
- 12 bit value is transferred within 30 ms (6 AS-Interface-cycles)

Gateway for Fieldbusses

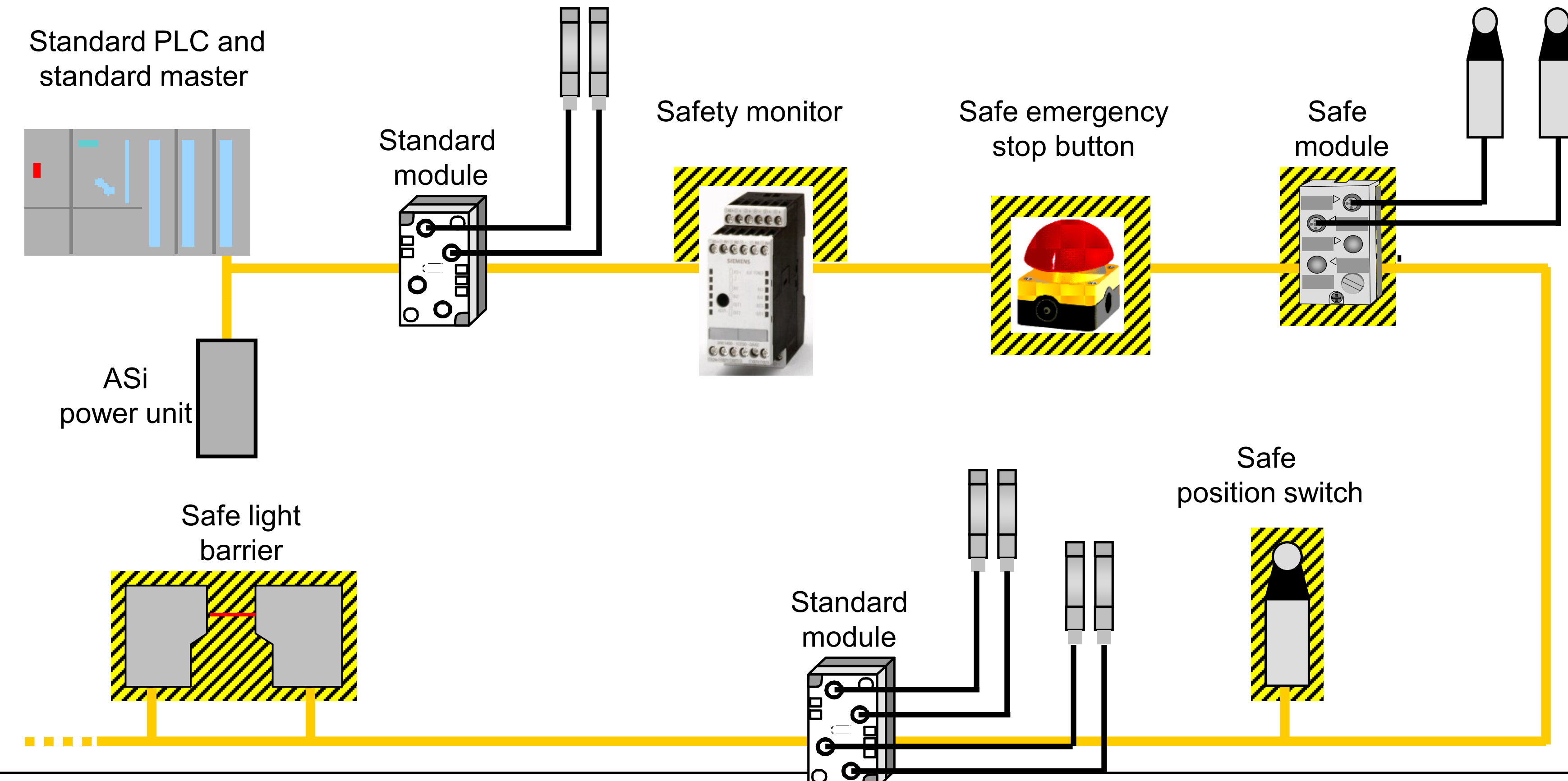


AS-i Safety at Work

Actuator-Sensor-Interface

SAFETY AT WORK

Safety-related and standard components in an AS-Interface network



Users Guideline (GB)

AS-International Association

The System with Safety

- Integration of all binary safety-related components in AS-Interface, such as:
  - Emergency stop
  - Protective door switch
  - Safety light barriers/grids
  - Contactor controls, etc.
- Mixed operation of standard and safety-related components via the yellow AS-i cable
- Fail-safe PLC or special master not necessary due to safety monitor
- It is possible to form groups of safe signals within a network
- Diagnostics via standard master/standard PLC

Features of Safety at Work

- Can be used up to control category 4 according to EN 954 -1
- Response time max. 35 ms
- Approved by BIA[1] and TÜV[2]
- Full compatibility with all other AS components according to EN 50 295
- Use of the standard AS-Interface protocol
- Maximum extension of the system with 31 safe slaves
- Stop category 0 or 1 can be set

BIA [1]: Berufsgenossenschaftliches Institut für Arbeitssicherheit

TÜV [2]: Technischer Überwachungsverein

First steps ...

1. Addressing the slaves

In one AS-Interface® network the address area is 01 up to 31

Slaves have always the address 00 when delivered.

Double addressing of the slaves is forbidden

Address slaves via the automatic mode of a master or via a addressing device

Label the slaves !

Program all addresses of all slaves in an AS-Interface® network

2. Installation

Install the lower parts of the modules or of the slaves first

Place profiled cable AS-Interface® into right position

If needed use also the cable for auxiliary supply

Screw tight the upper part of the module

Connect sensors and / or actuators via standard cable

First steps ...

3. Commissioning phase

Check properly the correct connection of the AS-i supply and the auxiliary supply (yellow and if needed black cable in right position ?)

Switch on the AS-Interface master

The master recognises all slaves during the commissioning phase and compares with a project list.

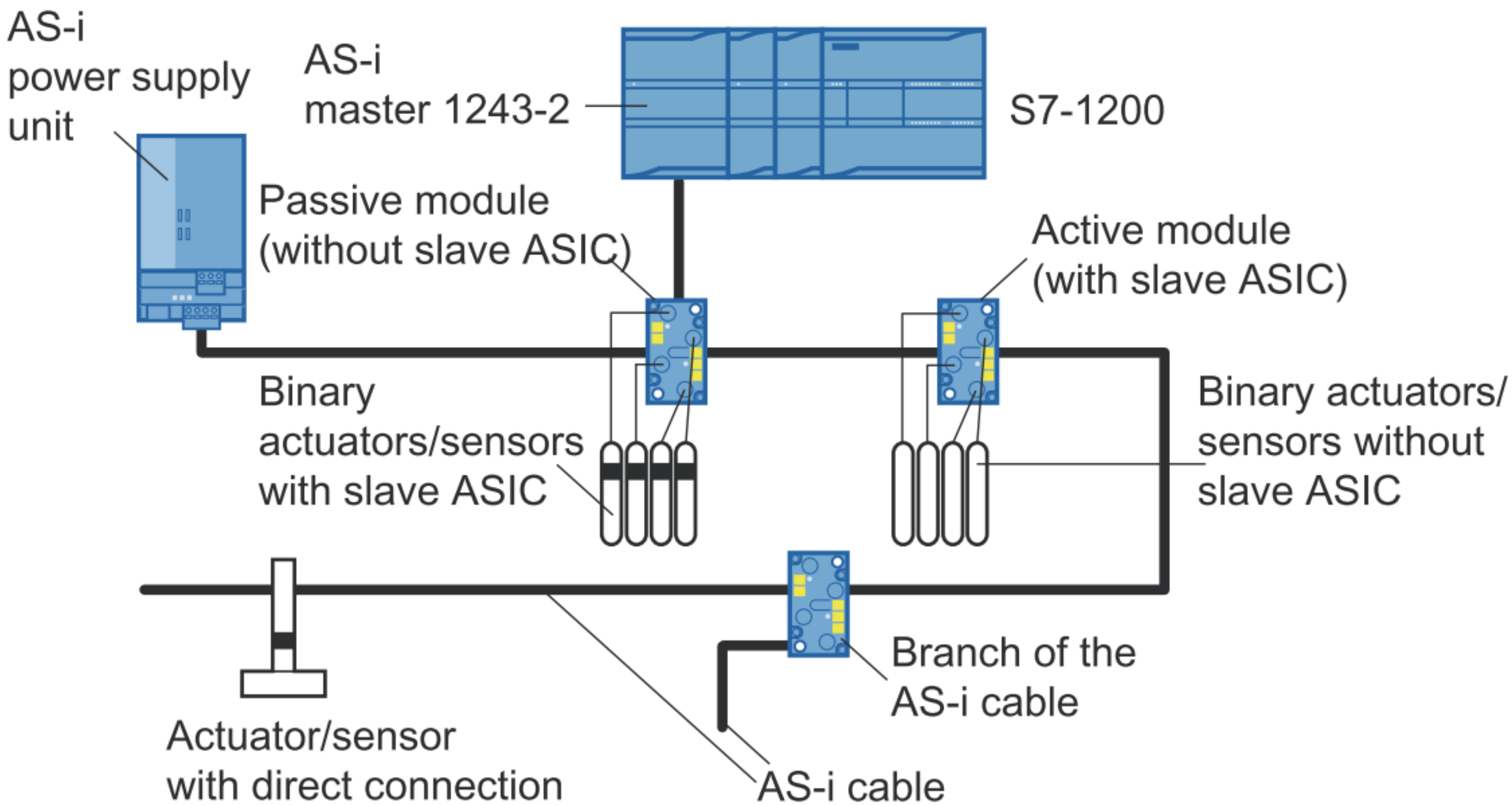
In case of an error a configuration error bit is set

Start the PLC program to control the AS-Interface ® application

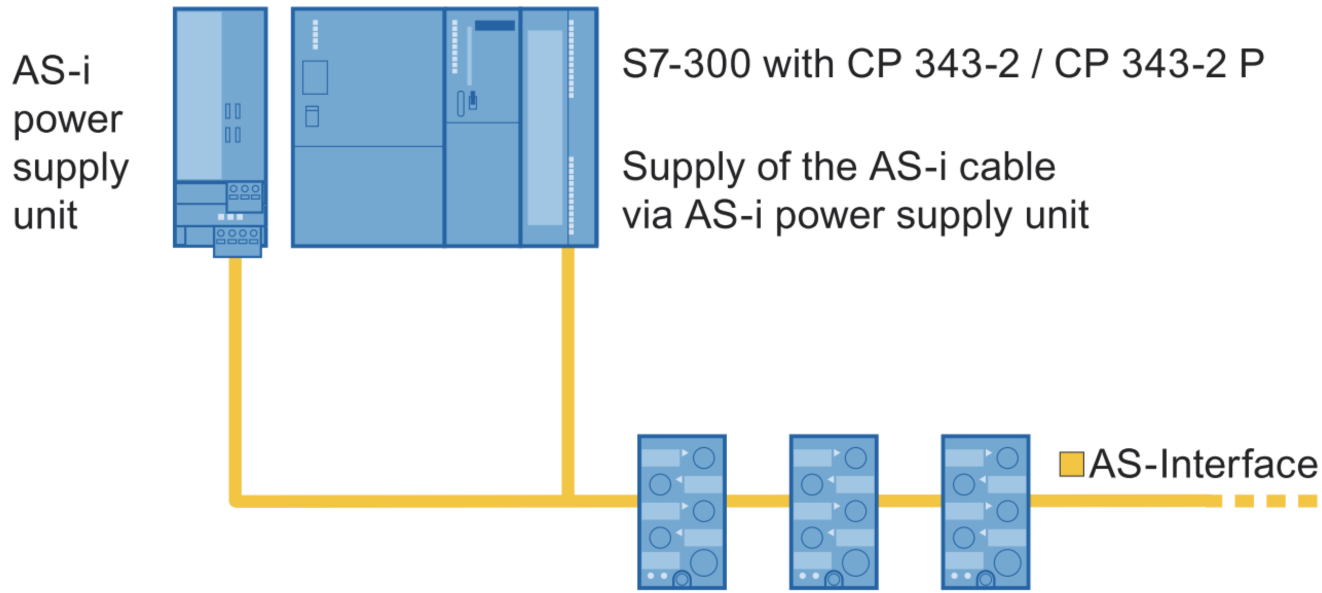
Note:

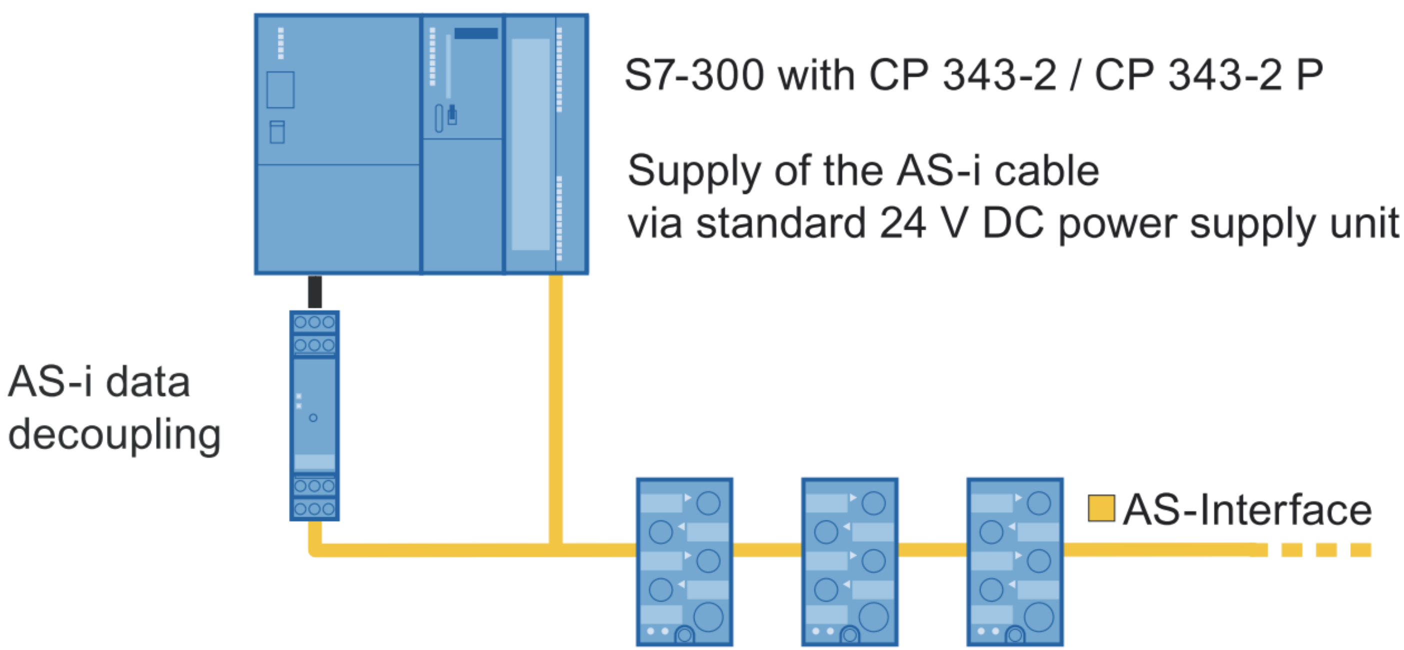
- It is possible to check the inputs and outputs without a PLC

- Functional units can put into service independent of other parts of a plant



AS-i
power supply
unit
AS-i
master 1243-2
S7-1200
Passive module
(without slave ASIC)
Active module
(with slave ASIC)
Binary
actuators/sensors
with slave ASIC
Binary actuators/
sensors without
slave ASIC
Branch of the
AS-i cable
Actuator/sensor
with direct connection
AS-i cable


AS-i
power
supply
unit
S7-300 with CP 343-2 / CP 343-2 P
Supply of the AS-i cable
via AS-i power supply unit
AS-Interface


S7-300 with CP 343-2 / CP 343-2 P
Supply of the AS-i cable
via standard 24 V DC power supply unit
AS-i data
decoupling
AS-Interface

AS-Interface – Components

- Gateways to industrial buses
- Backplane scanners for dominant PLCs

PROFI BUS
PROCESS FIELD BUS
INDUSTRIAL ETHERNET
DeviceNet

PEPPERL+FUCHS

AS-Interface – Components

Repeater

- Allows larger networks
- IP20 or IP67
- Selectable termination (IP20)

Power Conditioner

- Standard DC supply powers AS-Interface
- Dual-network model powers two networks

Application hint

- Design a network using an actual AS-Interface power supply
- Observe the "100 m segment rule"
- Contact Pepperl+Fuchs before violating the "100 m segment rule" or using a power conditioner

AS-Interface – Components

I/O Modules

- Sensors and actuators are connected to I/O modules
- Inputs (sensors) are powered from AS-Interface
- Outputs are powered from AUX power supply

IP20
Enclosure-mounted
applications

≥IP65
Field-mounted
applications

PEPPERL+FUCHS

AS-Interface – Components

Analog modules

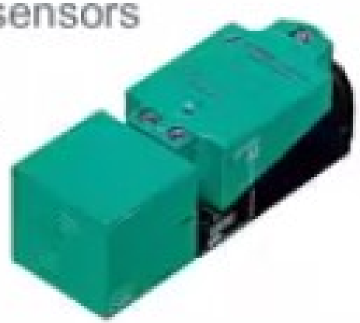
- Voltage/current/temperature
- Input and output

Encoders

- Absolute multiturn and single-turn

Advanced diagnostics sensors

- Coil monitoring
- Precollision indication
- Dirty-lens detection



Application hint

- Try solving the application using conventional sensors first

AS-Interface – Components

Stack Lights

- Stack Lights
- Color modules
- Sound modules
- Several mounting options
- Can also be used in non-AS-Interface installations

Specialty Modules

- Pushbuttons
- AC inputs
- Pneumatic modules
- Valve position sensors

PEPPERL+FUCHS

Summary of This Section

An AS-Interface system needs

- A gateway or scanner card to interface with the PLC
- An AS-Interface power supply to power
- I/O modules
- Sensors connected to modules that are connected to the network segments
- Network cable and accessories

IO-Link

SENSOP

Some Key Benefits

- Uses Twist-on connectors instead of wire terminations.
- More Diagnostics.
- Change setpoints from PLC.
- Compatible with existing sensors.

SENSOPART

IEC 61131 – Standard for Programmable Controllers

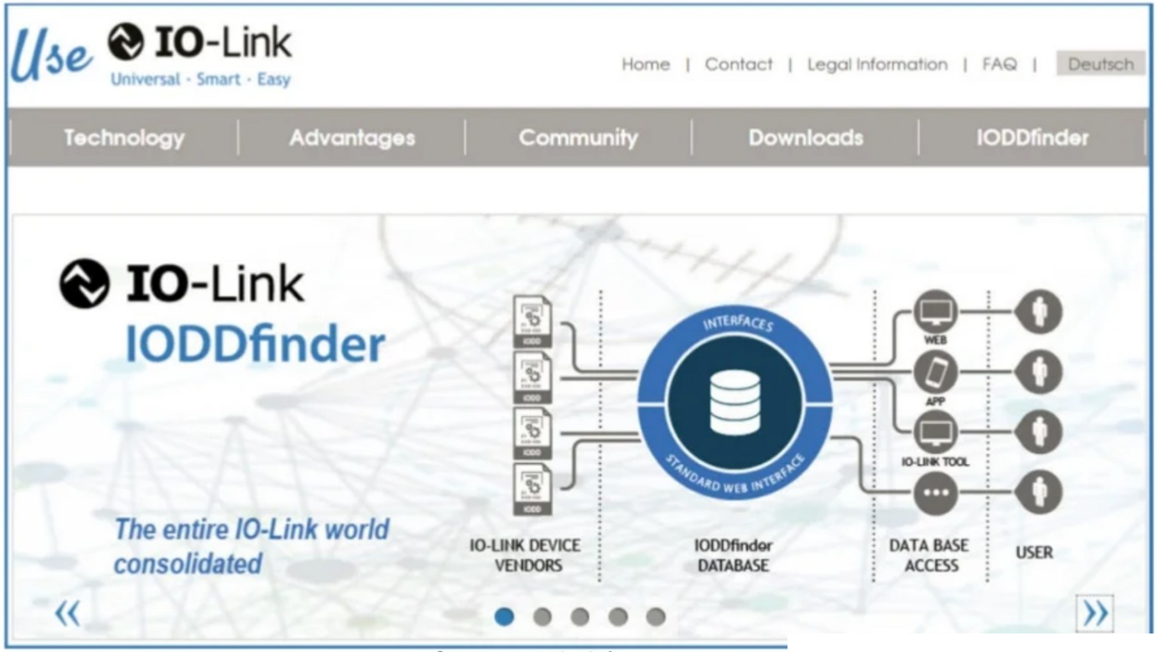
61131-3 – PLC Languages.

- Ladder
- Function Block
- Statement List
- Structured Text
- Sequential Function Charts

61131-9 - Single-drop digital communication interface for small sensors and actuators (SDCI).

- This is marketed as IO-Link.

www.io-link.com



from www.IO-Link.com

Some IO-Link Member companies

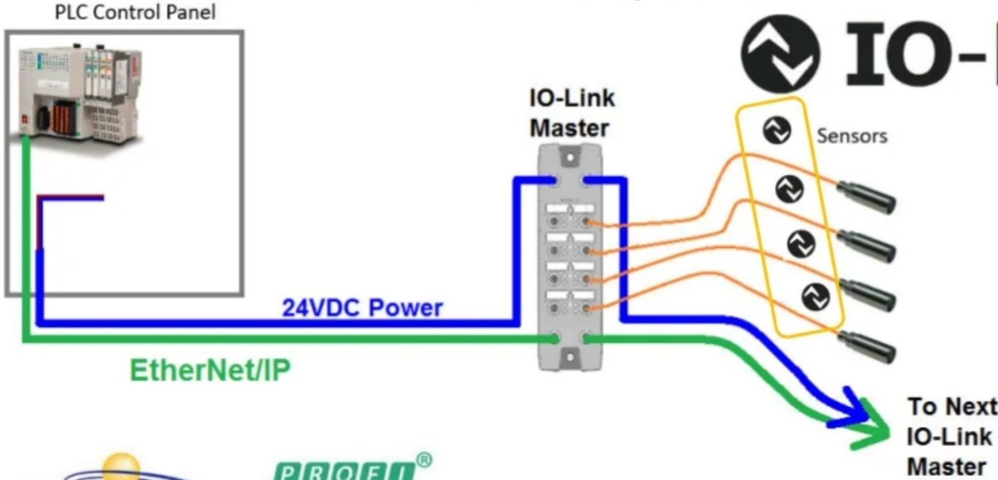
from www.IO-Link.com

dr hab.

SENSOPART

An IO-Link System

IO-Link



EtherNet/IP PROFI NET

SENSOPART

Components of An IO-Link System

PLC

IO-Link Master

Cable

IO-Link Sensor

IODD File

PC

EtherNet/IP PROFI NET EtherCAT

IO-Link

The Reality:

Any PLC*

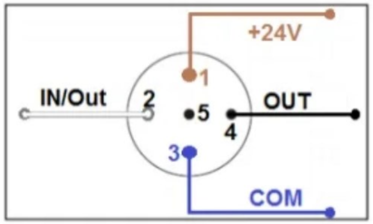
Any Sensor**

No Compatibility Matrix

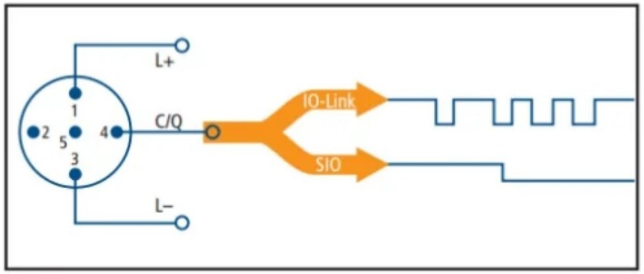
Any Master

Sensor Reality:

Remote I/O M12:



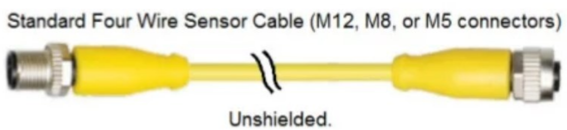
IO-Link Master M12:



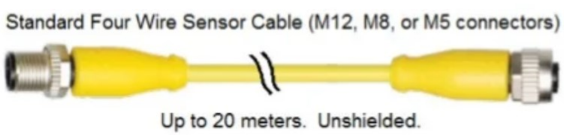
IO-Link Masters recognize both
IO-Link sensors and
24VDC sensors.

Wiring:

Traditional Sensor:



IO-Link Sensor:



Connectors are the same M12/M8/M5 as always.

Images from www.balluff.com

Next: Machine Wiring Evolution

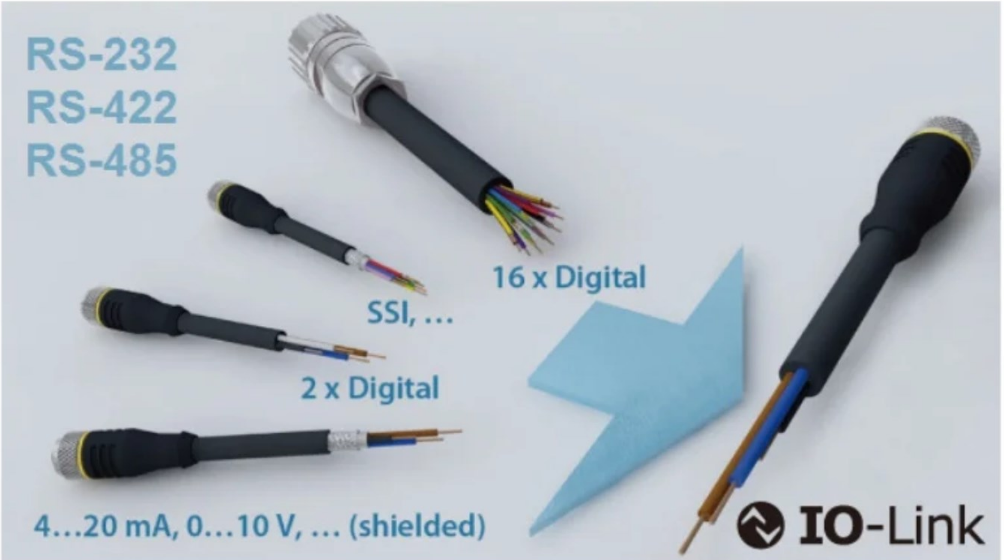
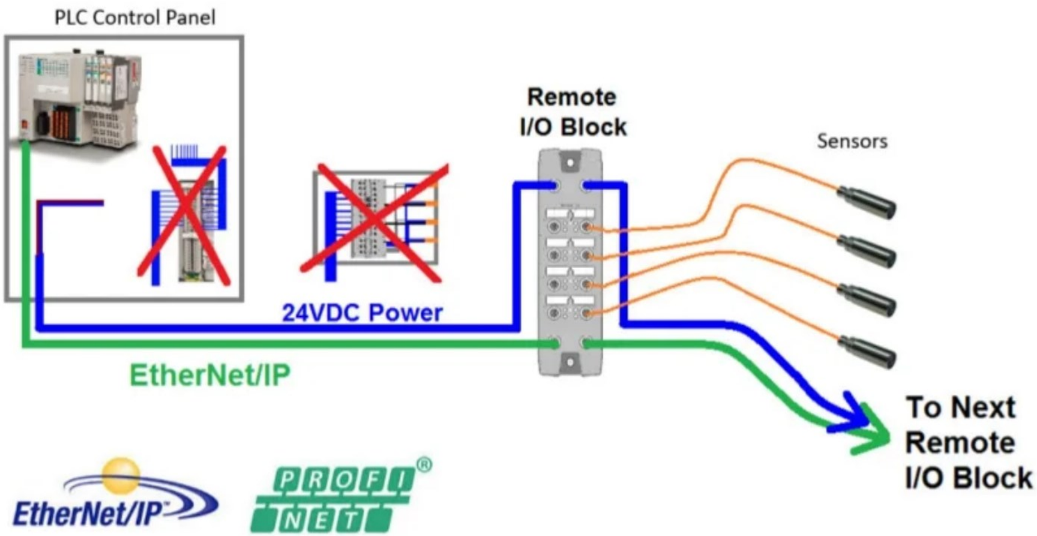
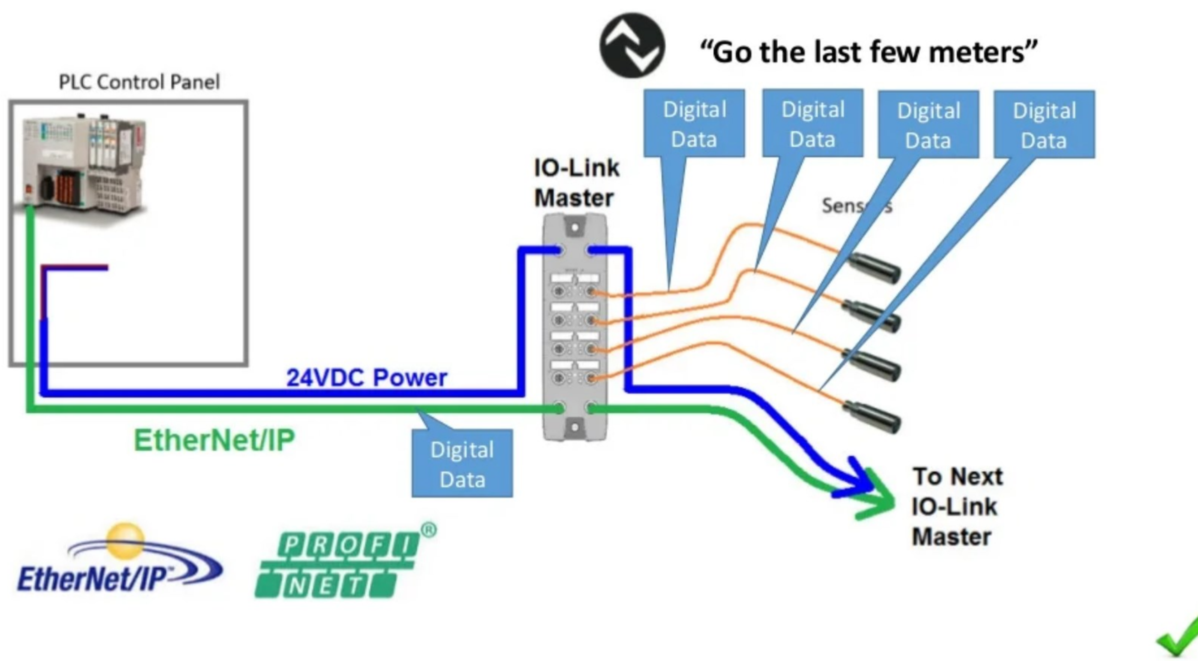


Image By Simon Muckenhirn / Violette Koppe

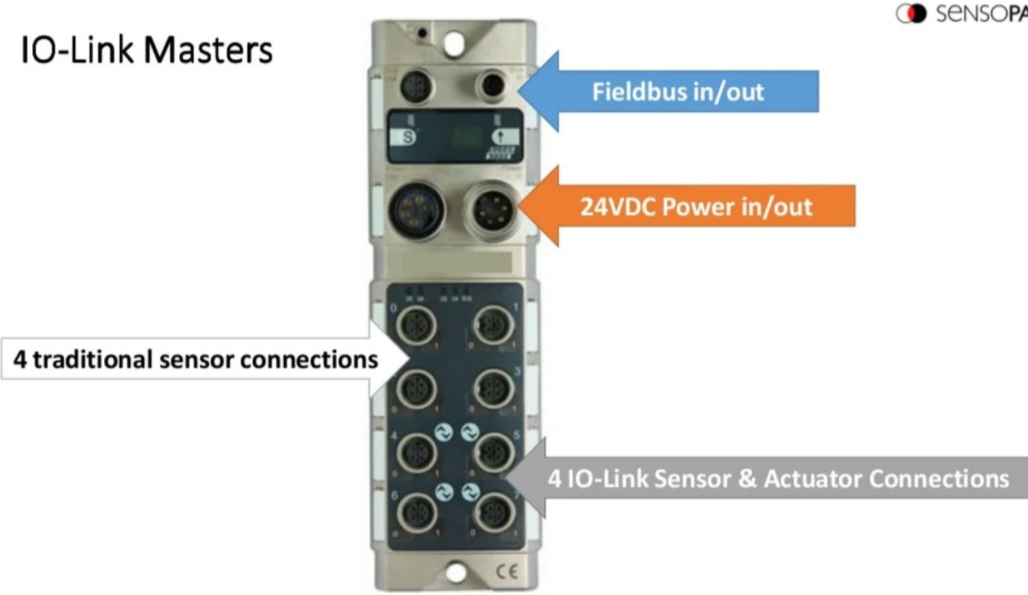
Fieldbus & Remote I/O Blocks



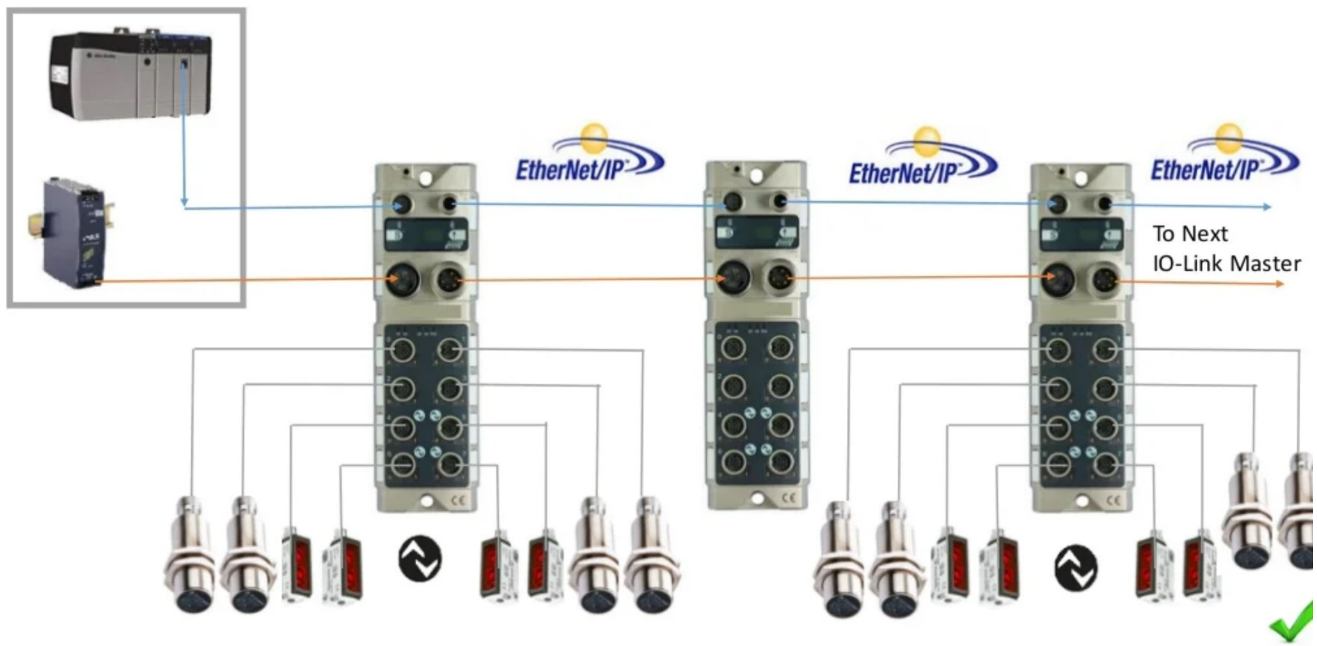
IO-Link Master



IO-Link Masters



IO-Link Masters



SensoPart

Next: Sensor Evolution

Image from http://mindgasms.theblogpress.com

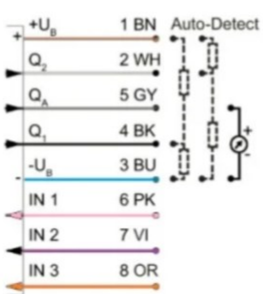
Images from www.SensoPart.com

Typical Discrete Sensor - 1990

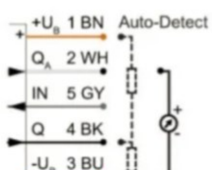
IO-Link

- Multiple Discrete Outputs.
  - Weak signal.
  - Multiple Setpoints.
- Multiple Discrete Inputs.
  - Remote teach.
  - Multiple configurations.

Typical Sensor – 21st Century



8-pin



5-pin

Pin	Description
A	Transmitter data / SSI: Data +
B	Signal output Q1
C	Receiver data / SSI: Clock +
D	Analogue output
E	Service output Qs
F	Plausibility output Qp
G	Ub + 18 to 30 V
H	Receiver data / SSI: Clock -
J	Signal output Q
K	Transmitter data / SSI: Data -
L	Signal output Q2
M	0 V (GND)

16-pin

Analog Signals

- 4-20mA or 0-10 VDC.
- 12 bit A/D converters.
- Best Resolution = 2^12 = 4096
- Sensor Example
  - Range of 70 meters
  - Resolution of 2 millimeters.
  - 70 / (2/1000) = 35,000 increments.
  - With an A/D it's only 4096 (8 x)

Resolution is lost just
by using an analog input

IO-Link

Configure Double-Integer register
In the PLC to be updated repeatedly.

Resolution is preserved

IODD File

- IODD = Input / Output Device Description.
  Like EDS or GSD
- XML Format.
- The IO-Link master has to have a copy of the IODD file for each unique sensor connected to it.

IODD files

Data Table (FT 55-CM - Color Senosr)

PROCESS DATA PROFILE 0: SWITCHING OUTPUTS

Byte	7	6	5	4	3	2	1	0
Byte 0								
Byte 1								
Byte 2	Quality MSB D7	D6	D5	D4	D3	D2	D1	Quality LSB D0
Byte 3								Q quality
Byte 4								
Byte 5		Switching output Q7	Switching output Q6	Switching output Q5	Switching output Q4	Switching output Q3	Switching output Q2	Switching output Q1

Signal quality level [%]
Q quality, threshold adjustable [0/1]
Switching outputs [0/1]

PROCESS DATA PROFILE 1: COLOR VALUES

Byte	7	6	5	4	3	2	1	0
Byte 0	Energy MSB D11	D10	D9	D8	D7	D6	D5	D4
Byte 1	D3	D2	D1	Energy LSB D0	Blue MSB D11	D10	D9	D8
Byte 2	D7	D6	D5	D4	D3	D2	D1	Blue LSB D0
Byte 3	Green MSB D11	D10	D9	D8	D7	D6	D5	D4
Byte 4	D3	D2	D1	Green LSB D0	Red MSB D11	D10	D9	D8
Byte 5	D7	D6	D5	D4	D3	D2	D1	Red LSB D0

Energy value [%]
Color value blue [%]
Color value green [%]
Color value red [%]

Types of data & speeds

- **Process data -** Sent Cyclically (every communication cycle)

Examples are Distance Measured, and Part Present.

About every 2.3 ms. Can contain between 1 bit and 32 bytes of information.

- **Service data -**Service protocol data units (SPDUs) allow the user to retrieve detailed information about the device.

Up to 16,000 blocks. ~400ms.

Examples include configuration data (e.g. activation, deactivation of functions).

And Sensor manufacturer no. & model no.

- **Events -** Events are reported without waiting for an SPDU to be queried.

Examples include diagnostics.

• COM1 - 4,800 baud • COM2 - 38,400 baud • COM3 230,400 baud